

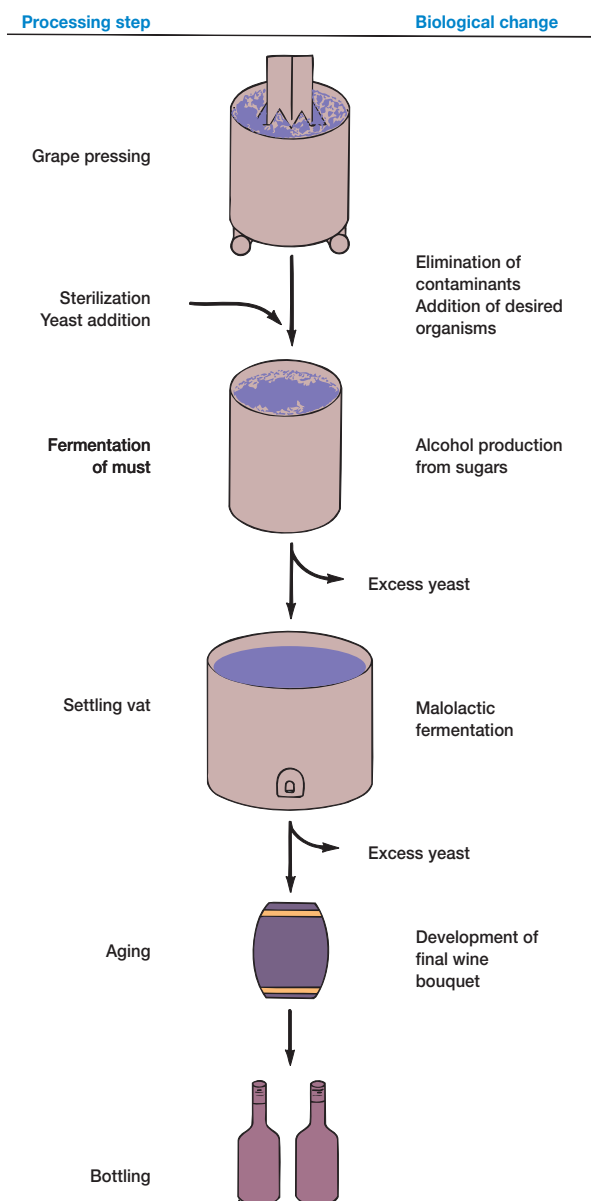


Figure 41.19 Wine Making. Once grapes are pressed, the sugars in the juice (the must) can be immediately fermented to produce wine. Must preparation, fermentation, and aging are critical steps.

Many processing variations can be used during wine production. The wine can be distilled to make a “burned wine” or brandy. *Acetobacter* and *Gluconobacter* can be allowed to oxidize the ethanol to acetic acid and form a **wine vinegar**. In the past an acetic acid generator was used to recirculate the wine over a bed of wood chips, where the desired microorganisms developed as a surface growth. Today the process is carried out in large aerobic submerged cultures under much more controlled conditions.

Natural champagnes are produced by continuing the fermentation in bottles to produce a naturally sparkling wine. Sediments that remain are collected in the necks of inverted champagne bottles after the bottles have been carefully turned. The necks of the bottles are then frozen and the corks removed to disgorge the accumulated sediments. The bottles are refilled with clear champagne from another disgorged bottle, and the product is ready for final packaging and labeling.

Beers and Ales

Beer and ale production uses cereal grains such as barley, wheat, and rice. The complex starches and proteins in these grains must be changed to a more readily usable mixture of simpler carbohydrates and amino acids. This process, shown in **figure 41.20**, involves germination of the barley grains and activation of their enzymes to produce a **malt**. The malt is then mixed with water and the desired grains, and the mixture is transferred to the mash tun or cask in order to hydrolyze the starch to usable carbohydrates. Once this process is completed, the **mash** is heated with hops (dried flowers of the female vine *Humulus lupulus*), which were originally added to the mash to inhibit spoilage microorganisms. The hops also provide flavor and assist in clarification of the wort. In this heating step the hydrolytic enzymes are inactivated and the wort can be **pitched**—inoculated—with the desired yeast.

Most beers are fermented with **bottom yeasts**, related to *Saccharomyces carlsbergensis*, which settle at the bottom of the fermentation vat. The beer flavor also is influenced by the production of small amounts of glycerol and acetic acid. Bottom yeasts produce beer with a pH of 4.1 to 4.2 and requiring 7 to 12 days of fermentation (**figure 41.21**). With a top yeast, such as *Saccharomyces cerevisiae*, the pH is lowered to 3.8 to produce ales. Freshly fermented (green) beers are aged or **lagered**, and when they are bottled, CO₂ is usually added. Beer can be pasteurized at 140°F or higher or sterilized by passage through membrane filters to minimize flavor changes.

In many places there is increased interest in specialty beers. Local “Braumeisters” develop unique products with special brewing techniques and ingredients. Often unusual names are given to these creations. This has led to local tasting festivals and increased interest in the art and craft of brewing.

Distilled Spirits

Distilled spirits are produced by an extension of beer production processes. The fermented liquid is boiled, and the volatile components are condensed to yield a product with a higher alcohol content than beer. Rye and bourbon are examples of whiskeys. Rye whiskey must contain at least 51% rye grain, and bourbon must contain at least 51% corn. Scotch whiskey is made primarily of barley. Usually a **sour mash** is used; the mash is inoculated with a homolactic (lactic acid is the major fermentation product) bacterium such as *Lactobacillus delbrueckii*, which can lower the mash pH to around 3.8 in 6 to 10 hours. This limits the development of undesirable organisms. Vodka and grain alcohols are also produced by distillation. Gin is vodka to which resinous flavoring

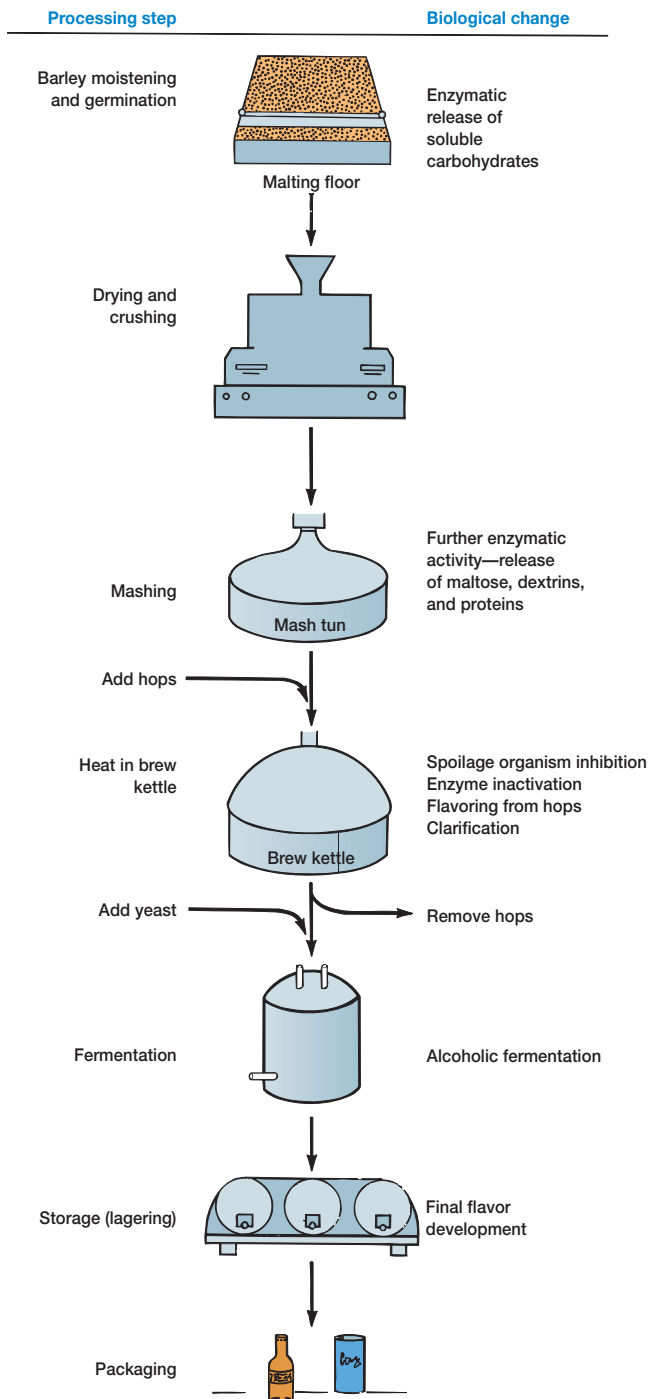


Figure 41.20 Producing Beer. To make beer, the complex carbohydrates in the grain must first be transformed into a fermentable substrate. Beer production thus requires the important steps of malting, and the use of hops and boiling for clarification, flavor development, and inactivation of malting enzymes. Only after completion of these steps can the actual fermentation be carried out.



Figure 41.21 Beer Fermentation. Beer making can be carried out on a large scale or in the kitchen. In large-scale processes, copper brewing vats are often used, as shown here in the Carlsberg Brewery, Copenhagen, Denmark.

agents—often juniper berries—have been added to provide a unique aroma and flavor.

Production of Breads

Bread is one of the most ancient of human foods, and is produced with the help of microorganisms. The use of yeasts to leaven bread is carefully depicted in paintings from ancient Egypt. A bakery at the Giza Pyramid area, from the year 2575 B.C., has been excavated. It is estimated that 30,000 people a day were provided with bread from this bakery. Samples of bread from 2100 B.C. are on display in the British Museum. In breadmaking, yeast growth is carried out under aerobic conditions. This results in increased CO_2 production and minimum alcohol accumulation. The fermentation of bread involves several steps: alpha- and beta-amylases present in the moistened dough release maltose and sucrose from starch. Then a baker's strain of the yeast *Saccharomyces cerevisiae*, which contains maltase, invertase, and zymase enzymes, is added. The CO_2 produced by the yeast results in the light texture of many breads, and traces of fermentation products contribute to the final flavor. Usually bakers add sufficient yeast to allow the bread to rise within 2 hours—the longer the rising time, the more additional growth by contaminating bacteria and fungi can occur, making the product less desirable.

By using more complex assemblages of microorganisms, bakers can produce special breads such as sour doughs. The yeast *Saccharomyces exiguus*, together with a *Lactobacillus* species, produces the characteristic acidic flavor and aroma of such breads.

Bread products can be spoiled by *Bacillus* species that produce ropiness. If the dough is baked after these organisms have grown, stringy and ropy bread will result, leading to decreased consumer acceptance.

Table 41.9 Fermented Foods Produced from Fruits, Vegetables, Beans, and Related Substrates

Foods	Raw Ingredients	Fermenting Microorganisms	Area
Coffee	Coffee beans	<i>Erwinia dissolvans</i> , <i>Saccharomyces</i> spp.	Brazil, Congo, Hawaii, India
Gari	Cassava	<i>Corynebacterium manihoti</i> , <i>Geotrichum</i> spp.	West Africa
Kenkey	Corn	<i>Aspergillus</i> spp., <i>Penicillium</i> spp., Lactobacilli, yeasts	Ghana, Nigeria
Kimchi	Cabbage and other vegetables	Lactic acid bacteria	Korea
Miso	Soybeans	<i>Aspergillus oryzae</i> , <i>Zygosaccharomyces rouxii</i>	Japan
Ogi	Corn	<i>Lactobacillus plantarum</i> , <i>Lactococcus lactis</i> , <i>Zygosaccharomyces rouxii</i>	Nigeria
Olives	Green olives	<i>Leuconostoc mesenteroides</i> , <i>Lactobacillus plantarum</i>	Worldwide
Ontjom	Peanut presscake	<i>Neurospora sitophila</i>	Indonesia
Peujeum	Cassava	Molds	Indonesia
Pickles	Cucumbers	<i>Pediococcus cerevisiae</i> , <i>L. plantarum</i>	Worldwide
Poi	Taro roots	Lactic acid bacteria	Hawaii
Sauerkraut	Cabbage	<i>L. mesenteroides</i> , <i>L. plantarum</i>	Worldwide
Soy sauce	Soybeans	<i>Aspergillus oryzae</i> or <i>A. soyae</i> , <i>Z. rouxii</i> , <i>Lactobacillus delbrueckii</i>	Japan
Sufu	Soybeans	<i>Mucor</i> spp.	China
Tao-si	Soybeans	<i>A. oryzae</i>	Philippines
Tempeh	Soybeans	<i>Rhizopus oligosporus</i> , <i>R. oryzae</i>	Indonesia, New Guinea, Surinam

Adapted from James M. Jay. 2000. *Modern Food Microbiology*, 6th edition. Reprinted by permission of Aspen Publishing, Frederick, Md.

Other Fermented Foods

Many other plant products can be fermented, as summarized in **table 41.9**. These include sufu, which is produced by the fermentation of tofu, a chemically coagulated soybean milk product. To carry out the fermentation, the tofu curd is cut into small chunks and dipped into a solution of salt and citric acid. After the cubes are heated to pasteurize their surfaces, *Actinimucor elegans* and other *Mucor* species are added. When a white mycelium develops, the cubes, now called pehtze, are aged in salted rice wine. This product has achieved the status of a delicacy in many parts of the Western world. Another popular product is tempeh, a soybean mash fermented by *Rhizopus* (**figure 41.22**).

Sauerkraut or sour cabbage, is produced from wilted, shredded cabbage, as shown in **figure 41.23**. Usually the mixed microbial community of the cabbage is used. A concentration of 2.2 to 2.8% sodium chloride restricts the growth of gram-negative bacteria while favoring the development of the lactic acid bacteria. The primary microorganisms contributing to this product are *Leuconostoc mesenteroides* and *Lactobacillus plantarum*. A predictable microbial succession occurs in sauerkraut's development. The activities of the lactic acid-producing cocci usually cease when the acid content reaches 0.7 to 1.0%. At this point *Lactobacillus plantarum* and *Lactobacillus brevis* continue to function. The final acidity is generally 1.6 to 1.8, with lactic acid comprising 1.0 to 1.3% of the total acid in a satisfactory product.

Pickles are produced by placing cucumbers and such components as dill seeds in casks filled with a brine. The sodium chloride concentration begins at 5% and rises to about 16% in 6 to 9 weeks. The salt not only inhibits the growth of undesirable bacte-

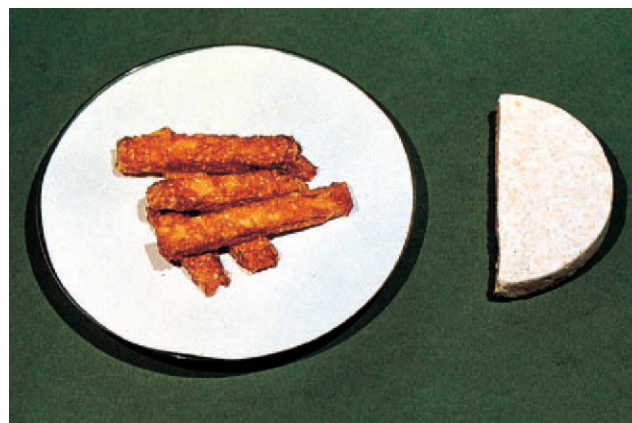


Figure 41.22 Fermented Soybean Products. Microorganisms also can be used to process soybean products. Tempeh is produced from soybeans and *Rhizopus*. Both the raw cake and the fried food are shown.

ria but also extracts water and water-soluble constituents from the cucumbers. These soluble carbohydrates are converted to lactic acid. The fermentation, which can require 10 to 12 days, involves the development of *Leuconostoc mesenteroides*, *Enterococcus faecalis*, *Pediococcus cerevisiae*, *Lactobacillus brevis*, and *L. plantarum*. *L. plantarum* plays the dominant role in this fermentation process. Sometimes, to achieve more uniform pickle quality, natural microorganisms are first destroyed and the cucumbers are fermented using pure cultures of *P. cerevisiae* and *L. plantarum*.

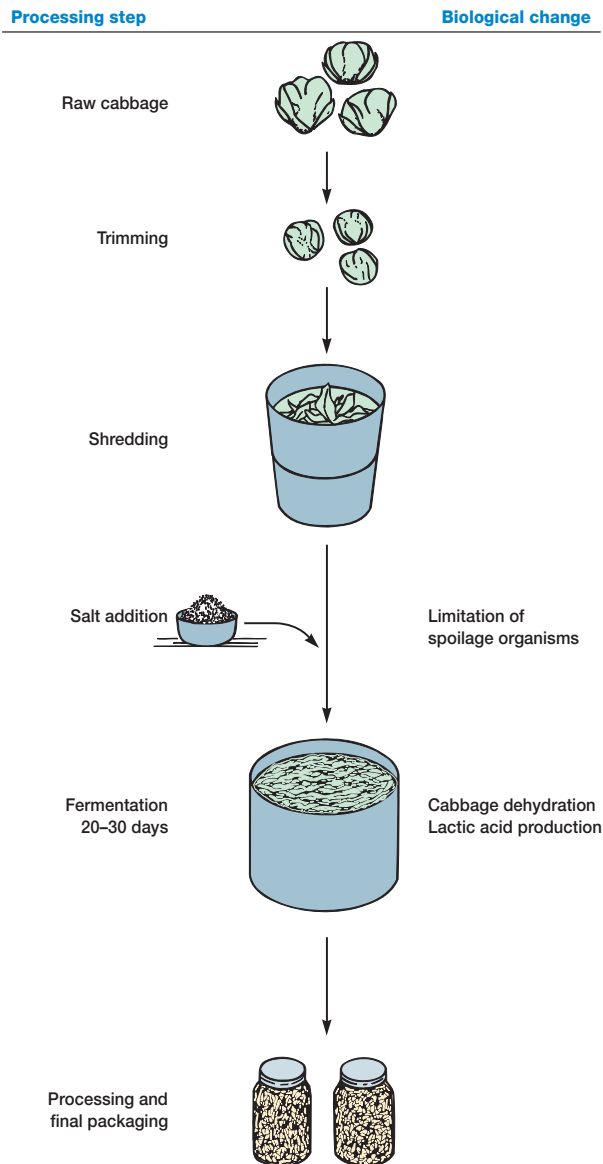


Figure 41.23 Sauerkraut. Sauerkraut production employs a lactic acid fermentation. The basic process involves fermentation of shredded cabbage in the presence of 2.25–2.5% by weight of salt to inhibit spoilage organisms.

Grass, chopped corn, and other fresh animal feeds, if stored under moist anaerobic conditions, will undergo a lactic-type mixed fermentation that produces pleasant-smelling **silages**. Trenches or more traditional vertical steel or concrete silos are used to store the silage. The accumulation of organic acids in silage can cause rapid deterioration of these silos. Older wooden stave silos, if not properly maintained, allow the outer portions of the silage to become aerobic, resulting in spoilage of a large portion of the plant material.



Figure 41.24 Mushroom Farming. Growing mushrooms requires careful preparation of the growth medium and control of environmental conditions. The mushroom bed is a carefully developed compost, which can be steam sterilized to improve mushroom growth.

1. Describe and contrast the processes of wine and beer production. How are red wines produced when the juice of all grapes is white?
2. How do champagnes differ from wines?
3. Describe how distilled spirits like whiskey are produced.
4. How are bread, sauerkraut, and pickles produced? What microorganisms are most important in these fermentations?

41.7 Microorganisms as Foods and Food Amendments

Besides microorganisms' actions in fermentation as agents of physical and biological change, they themselves can be used as a food source. A variety of bacteria, yeasts, and other fungi have been used as animal and human food sources. Mushrooms (*Agaricus bisporus*) are one of the most important fungi used directly as a food source. Large caves are used to maintain optimal conditions for the production of this delicacy (**figure 41.24**). Microorganisms can be used directly as a food source or as a supplement to other foods and are then called single-cell protein. One of the more popular microbial food supplements is the cyanobacterium *Spirulina*. It is used as a food source in Africa and is now being sold in United States health food stores as a dried cake or powdered product.

Microorganisms such as *Lactobacillus* and *Bifidobacterium* are being used in the rapidly developing area of **probiotics**, the addition of microorganisms to the diet in order to provide health benefits beyond basic nutritive value. These supplements may be called **microbial dietary adjuvants**. Many of the claims for the value of probiotics in the past were not based on strong scientific investigations, and were more the stuff of tradition and hearsay. This is changing, as recent work in this area is based on a more

scientific foundation. For example, scientists have created a Simulated Human Intestinal Microbial Ecosystem or **SHIME system**, to better understand the effects of adding probiotic organisms. [Probiotics \(Box 31.1\)](#)

In addition, there is a greater appreciation of the role of oligosaccharide polymers or **prebiotics**, which are not processed until they enter the large intestine. The combination of prebiotics and probiotic microorganisms is described as a **synbiotic** system. This synbiotic combination can result in an increase in the levels of butyric and propionic acids, as well as an increase in *Bifidobacterium* in the human intestine. The butyrate, especially, may play a role in the possible beneficial effects of probiotics on intestinal processes.

Probiotics are being used successfully with poultry. Recently, the USDA has designated a probiotic *Bacillus* strain for use with chickens as “Generally Recognized As Safe” or GRAS. Feeding chickens a strain of *Bacillus subtilis* (Calsporin) led to increased body weight and feed conversion. There was also a reduction in coliforms and *Campylobacter* in the processed carcasses. It has been suggested that this probiotic decreases the

need for antibiotics in poultry production and pathogen levels on farms. *Salmonella* can be controlled by spraying a patented blend of 29 bacteria, isolated from the chicken cecum, on day-old chickens. As they preen themselves, the chicks ingest the bacterial mixture, establishing a functional microbial community in the cecum and limiting *Salmonella* colonization of the gut in a process called **competitive exclusion**. In 1998, this product, called PREEMPT™, was approved for use in the United States by the Food and Drug Administration.

1. What conditions are needed to have most efficient production of edible mushrooms? What is the scientific name of the most important fungus used for this purpose?
2. A cyanobacterium is widely used as a food supplement. What is this genus, and in what part of the world was it first used as a significant food source?
3. What are prebiotics, probiotics, and synbiotics?
4. What probiotic has recently been recognized as “Generally Recognized As Safe” or GRAS?

Summary

1. Most foods, especially when raw, provide an excellent environment for microbial growth. This growth can lead to spoilage or preservation, depending on the microorganisms present and environmental conditions.
2. The course of microbial development in a food is influenced by the intrinsic characteristics of the food itself—pH, salt content, substrates present, water presence and availability—and extrinsic factors, including temperature, relative humidity, and atmospheric composition (**figure 41.1**).
3. Microorganisms can spoil meat, dairy products, fruits, vegetables, and canned goods in several ways. Spices, with their antimicrobial compounds, sometimes protect foods.
4. Modified atmosphere packaging (MAP) is being used to control microbial growth in foods and to extend product shelf life. This process involves decreasing oxygen and increased carbon dioxide levels in the space between the food surface and the wrapping material.
5. Food spoilage is a major concern throughout the world. This can occur at any point in the food production process: growth, harvesting, transport, storage, or final preparation. Spoilage also can occur if foods are not stored properly.
6. Fungi, if they can grow in foods, especially cereals and grains, can produce important disease-causing chemicals, including aflatoxins and fumonisins (carcinogens), and ergot alkaloids (mind-altering drugs).
7. Prior illness with hepatitis B can increase susceptibility to liver cancer from aflatoxins. Control of hepatitis B is suggested to be more critical than control of aflatoxins.
8. Algal toxins can be transmitted to humans through the food chain via marine products. These can have severe amnesic, diarrhetic and neurotoxic effects.
9. Foods can be preserved in a variety of physical and chemical ways, including filtration, alteration of temperature (cooling, pasteurization, sterilization), drying, the addition of chemicals, radiation, and fermentation (**table 41.4**).
10. There is an increasing interest in using bacteriocins for food preservation. Nisin, a product of *Streptococcus lactis*, is the major substance approved for use in foods. Bacteriocins are especially important for control of *Listeria monocytogenes*.
11. Foods can be contaminated by pathogens at any point in the food production, storage, or preparation processes. Pathogens such as *Salmonella*, *Campylobacter*, *Listeria*, and *E. coli* can be transmitted by the food to the susceptible consumer, where they grow and cause disease, or a food-borne infection (**table 41.6**). If the pathogen grows in the food before consumption and forms toxins that affect the food consumer without further microbial growth, the disease is a food-borne intoxication. Examples are intoxications caused by *Staphylococcus*, *Clostridium*, and *Bacillus*.
12. Norwalk-type viruses, *Campylobacter*, and *Salmonella* are thought to be the most important causes of food-borne illness. The cause of most food-borne illnesses is not known.
13. *E. coli* O157:H7 is an enterohemorrhagic bacterium that apparently can produce shigalike verocytotoxins, which especially affect the young. Proper food handling and thorough cooking are critical in control.
14. New variant Creutzfeldt-Jakob disease (vCJD) is of increasing worldwide concern as a possible food-borne infectious agent, which is related to the occurrence of “mad cow disease.” The major means of vCJD transmission between animals is the use of mammalian tissue in ruminant animal feeds. There are difficulties in detecting such prohibited animal products in ruminant feeds.
15. The increasing popularity of raw foods such as sprouts, seafood, and raspberries has provided new routes for disease transmission. Increases in the international shipment of fresh foods contribute to this problem. *Cyclospora* is a protozoan of concern, and the major source is contaminated waters.
16. Detection of food-borne pathogens is a major part of food microbiology. The use of immunological and molecular techniques such as probes, PCR, and pulsed field electrophoresis is making it possible to link disease occurrences to a common infection source (**figure 41.13**). PulseNet and InfoNet programs are being used to coordinate these control efforts.
17. Dairy products can be fermented to yield a wide variety of cultured milk products (**table 41.7**). These include mesophilic, therapeutic, thermophilic, lacto-ethanolic, and mold-lactic products.

18. Growth of lactic acid-forming bacteria, often with the additional use of rennin, can coagulate milk solids. These solids can be processed to yield a wide variety of cheeses, including soft unripened, soft ripened, semisoft, hard, and very hard types (**table 41.8** and **figure 41.17**). Both bacteria and fungi are used in these cheese production processes.
19. Wines are produced from pressed grapes and can be dry or sweet, depending on the level of free sugar that remains at the end of the alcoholic fermentation (**figure 41.19**). Champagne is produced when the fermentation, resulting in CO₂ formation, is continued in the bottle.
20. Beer and ale are produced from cereals and grains. The starches in these substrates are hydrolyzed, in the processes of malting and mashing, to produce a fermentable wort. *Saccharomyces cerevisiae* is a major yeast used in the production of beer and ale (**figure 41.20**).
21. Many plant products can be fermented with bacteria, yeasts, and fungi. Important products are breads, soy sauce, sufu, and tempeh (**table 41.9**). Sauerkraut and pickles are produced in a fermentation process in which natural populations of lactobacilli play a major role (**figure 41.23**).
22. Microorganisms themselves can serve as an important food source. Mushrooms (*Agaricus bisporus*) are one of the most important fungi used as a food source. *Spirulina*, a cyanobacterium, also is a popular food source sold in specialty stores.
23. Many microorganisms, including some of those used to ferment milks, can be used as food amendments or microbial dietary adjuvants. Microorganisms such as *Lactobacillus* and *Bifidobacterium*, termed probiotics, can be used with oligopolysaccharides, termed prebiotics, to yield synbiotics. Several types of probiotic microorganisms are being used successfully in poultry production.

Key Terms

aflatoxin 967	kefir 979	putrefaction 965
algal toxin 968	lager 983	racking 982
bacteriocin 972	malt 983	radappertization 972
bottom yeast 983	mash 983	SHIME system 987
canning 970	mashing 982	silage 986
competitive exclusion 987	microbial dietary adjuvant 986	spermosphere 974
enology 982	modified atmosphere packaging (MAP) 966	sour mash 983
ergotism 967	must 982	starter culture 978
extrinsic factor 964	osmophilic microorganism 965	synbiotic 987
food-borne infection 973	pasteurization 970	wine vinegar 983
food intoxication 975	pitching 983	wort 982
FoodNet 977	prebiotic 987	xerophilic microorganism 965
fumonisin 968	probe 976	yogurt 978
GRAS 972	probiotic 986	
intrinsic factor 964	PulseNet 977	

Questions for Thought and Review

1. You are going through a salad line in a cafeteria at the end of the day. Which types of foods would you tend to avoid, and why?
2. Why does the law only allow foods to be pasteurized a single time? What could be the consequences of repeated pasteurization?
3. Why is it recommended that frozen foods, once thawed, not be frozen again? Suggest some of the reasons for this important principle of food handling.
4. Hamburger is notorious for having relatively high populations of microorganisms. How can these populations be controlled in preparation and storage?
5. Coffee cream is now packaged in small-portion servings that can be held at room temperature for extended periods. How do you think this product is produced and packaged?
6. What factors might limit the application of DNA- and RNA-based molecular techniques to the detection of pathogens in foods? How might these problems be overcome?
7. Why were aflatoxins not discovered before the 1960s? Do you think this was the first time they had grown in a food product to cause disease?
8. What advantage might the shiga-like toxin give *E. coli* O157:H7? Can we expect to see other “new” pathogens appearing, and what should we do, if anything?
9. Because of the possible beneficial effects of probiotic microorganisms on the intestinal microflora, scientists are becoming increasingly interested in their use as dietary supplements. How might modern biotechnology be used to improve the effectiveness of these microorganisms?
10. “New” food-borne pathogens continue to be discovered. How much of this phenomenon do you think is due to new techniques for recognizing existing organisms rather than the actual appearance of new organisms? How would you sort out the answer to this question?
11. How might microbial ecology be used to better protect humans from food-borne diseases?

Critical Thinking Questions

1. Fresh lemon slices are often served with raw or steamed seafood (oysters, crab, shrimp). From a food microbiology perspective, provide an explanation for their being served. Are there other examples in either the cooking or the serving of foods that not only enhance flavor, but might have an antimicrobial strategy? Consider the example of marinades.
2. Keep a record of what you eat for a day or two. Determine if the food, beverages, and snacks you ate could have been produced (at any level) without the aid of microorganisms. Indicate at what level(s) microorganisms were deliberately used. Be sure to consider ingredients such as citric acid, which is produced at the industrial level by several species of fungi.
3. Colonization of a susceptible human is critical to food-borne disease microorganisms. How might it be possible to modify foods to decrease these attachment processes?

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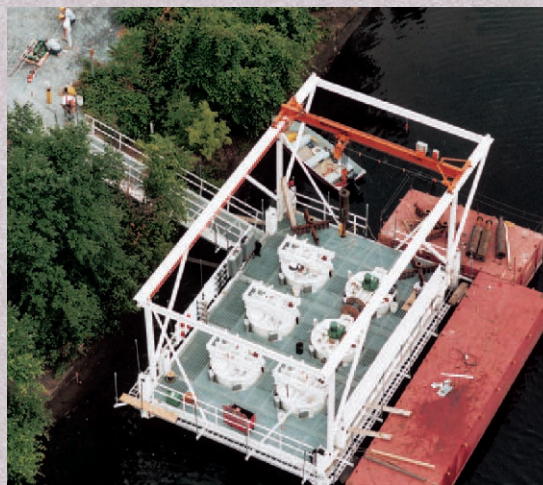
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CHAPTER 42

Industrial Microbiology and Biotechnology



Biodegradation often can be facilitated by changing environmental conditions. Polychlorinated biphenyls (PCBs) are widespread industrial contaminants that accumulate in anaerobic river muds. Although reductive dechlorination occurs under these anaerobic conditions, oxygen is required to complete the degradation process. In this experiment, muds are being aerated to allow the final biodegradation steps to occur.

Outline

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Concepts

1. Microorganisms are used in industrial microbiology and biotechnology to create a wide variety of products and to assist in maintaining and improving the environment.
2. Most work in industrial microbiology has been carried out using microorganisms isolated from nature or modified through mutations. In modern biotechnology, microorganisms with specific genetic characteristics can be constructed to meet desired objectives.
3. Most microorganisms have not been grown or described. A major challenge in biotechnology is to be able to grow and characterize these observed but uncultured microorganisms in what is called “bioprospecting.”
4. Forced evolution and adaptive mutations now are part of modern biotechnology. These can be carried out in processes termed “natural genetic engineering.”
5. The development of growth media and specific conditions for the growth of microorganisms is a large part of industrial microbiology and biotechnology. Microorganisms can be grown in controlled environments with specific limitations to maximize the synthesis of desired products.
6. Microbial growth in soils, waters, and other environments, where complex microbial communities already are present, cannot be completely controlled, and it is not possible to precisely define limiting factors or physical conditions.
7. Microbial growth in controlled environments is expensive; it is used to synthesize high-value microbial metabolites and other products for use in animal and human health. In comparison, microbial growth in natural environments usually does not involve the creation of specific microbial products; microorganisms are used to carry out lower-value environmental and agriculture-related processes.
8. In controlled growth systems, different products are synthesized during growth and after growth is completed. Most antibiotics are produced after the completion of active growth.
9. Antibiotics and other microbial products continue to contribute to animal and human welfare. Newer products include anticancer drugs. Combinatorial biology is making it possible to produce hybrid antibiotics with unique properties.
10. The products of industrial microbiology impact all aspects of our lives. These often are bulk chemicals that are used as food supplements and acidifying agents. Other products are used as biosurfactants and emulsifiers in a wide variety of applications.
11. Degradation is critical for understanding microbial contributions to natural environments. The chemical structure of substrates and microbial community characteristics play important roles in determining the fate of chemicals. Anaerobic degradation processes are important for the initial modification of many compounds, especially those with chlorine and other halogenated functions. Degradation can produce simpler or modified compounds that may not be less toxic than the original compound.
12. Biosensors are undergoing rapid development, which is limited only by the advances that are occurring in molecular biology and other areas of science. It is now possible, especially with streptavidin-biotin-linked systems, to have essentially real-time detection of important pathogens.
13. Gene arrays, based on recombinant DNA technology, allow gene expression to be monitored. These systems are being used in the analysis of complex microbial systems.
14. Bacteria, fungi, and viruses are increasingly employed as biopesticides, thus reducing dependence on chemical pesticides.
15. Application of microorganisms and their technology has both positive and negative aspects. Possible broader impacts of applications of industrial microbiology and biotechnology should be considered in the application of this rapidly expanding area.

Industrial microbiology and biotechnology both involve the use of microorganisms to achieve specific goals, whether creating new products with monetary value or improving the environment. Industrial microbiology, as it has traditionally developed, focuses on products such as pharmaceutical and medical compounds (antibiotics, hormones, transformed steroids), solvents, organic acids, chemical feedstocks, amino acids, and enzymes that have direct economic value. The microorganisms employed by industry have been isolated from nature, and in many cases, were modified using classic mutation-selection procedures.

The era of biotechnology has developed rapidly in the last several decades, and is characterized by the modification of microorganisms through the use of molecular biology, including the use of recombinant DNA technology (*see chapter 14*). It is now possible to manipulate genetic information and design products such as proteins, or to modify microbial gene expression. In addition, genetic information can be transferred between markedly different groups of organisms, such as between bacteria and plants.

Selection and use of microorganisms in industrial microbiology and biotechnology are challenging tasks that require a solid understanding of microorganism growth and manipulation, as well as microbial interactions with other organisms. The use of microorganisms in industrial microbiology and biotechnology follows a logical sequence. It is necessary first to identify or create a microorganism that carries out the desired process in the most efficient manner. This microorganism then is used, either in a controlled environment such as a fermenter or in complex systems such as in soils or waters to achieve specific goals.

42.1 Choosing Microorganisms for Industrial Microbiology and Biotechnology

The first task for an industrial microbiologist is to find a suitable microorganism for use in the desired process. A wide variety of alternative approaches are available, ranging from isolating microorganisms from the environment to using sophisticated molecular techniques to modify an existing microorganism.

Finding Microorganisms in Nature

Until relatively recently, the major sources of microbial cultures for use in industrial microbiology were natural materials such as soil samples, waters, and spoiled bread and fruit. Cultures from all areas of the world were examined in an attempt to identify strains with desirable characteristics. Interest in “hunting” for new microorganisms continues unabated.

Because only a minor portion of the microbial species in most environments has been isolated or cultured (**table 42.1**), there is a continuing effort throughout the world to find new microorganisms, even using environments that have been examined for decades. In spite of these long-term efforts, few microorganisms have been cultured and studied; most microbes

The microbe will have the last word.

—Louis Pasteur

Box 42.1

The Potential of Archaea from High-Temperature Environments for Use in Biotechnology

There is great interest in the characteristics of archaeans isolated from the outflow mixing regions above deep hydrothermal vents that release water at 250 to 350°C. This is because these hardy organisms can grow at temperatures as high as 113°C. The problems in growing these microorganisms are formidable. For example, to grow some of them, it will be necessary to use special culturing chambers and other specialized equipment to maintain water in the liquid state at these high temperatures.

Such microorganisms, termed hyperthermophiles, with optimum growth temperatures of 80°C or above (*see p. 126*), confront unique challenges in nutrient acquisition, metabolism, nucleic acid replication, and growth. Many of these are anaerobes that depend on elemental sulfur as

an oxidant and reduce it to sulfide. Enzyme stability is critical. Some DNA polymerases are inherently stable at 140°C, whereas many other enzymes are stabilized in vivo with unique thermoprotectants. When these enzymes are separated from their protectant, they lose their unique thermostability.

These enzymes may have important applications in methane production, metal leaching and recovery, and for use in immobilized enzyme systems. In addition, the possibility of selective stereochemical modification of compounds normally not in solution at lower temperatures may provide new routes for directed chemical syntheses. This is an exciting and expanding area of the modern biological sciences to which environmental microbiologists can make significant contributions.

Table 42.1 Estimated Total and Known Species from Different Microbial Groups

Group	Estimated Total Species	Known Species ^a	Percent Known
Viruses	130,000 ^b	5,000	[4] ^c
Archaea	?	<500	?
Bacteria	40,000 ^b	4,800	[12]
Fungi	1,500,000	69,000	5
Algae	60,000	40,000	67

^aMid-1990 values and should be increased 10–50%.

^bThese values are substantially underestimated, perhaps by 1–2 orders of magnitude.

^c[] indicates that these values are probably gross overestimates.

^dSmall subunit (SSU) rRNA data indicate much higher in situ diversity than given by culture-based studies.

Adapted from: D. A. Cowan. 2000. Microbial genomes—the untapped resource. *Tibtech* 18:14–16. Table 1, p. 15.

Table 42.2 Estimates of the Percent “Cultured” Microorganisms in Various Environments

Environment	Estimated Percent Cultured
Seawater	0.001–0.100
Fresh water	0.25
Mesotrophic lake	0.1–1.0
Unpolluted estuarine waters	0.1–3.0
Activated sludge	1–15
Sediments	0.25
Soil	0.3

Source: D. A. Cowan. 2000. Microbial genomes—the untapped resource. *Tibtech* 18:14–16. Table 2, p. 15.

that can be observed in nature have not been cultured or identified, although molecular techniques are making it possible to obtain information on these uncultured microorganisms (**table 42.2**). With increased interest in microbial diversity and microbial ecology, and especially in microorganisms from extreme environments (**Box 42.1**), microbiologists are rapidly expanding the pool of known microorganisms with characteristics desirable for use in industrial microbiology and biotechnology. They are also identifying microorganisms involved in mutualistic and protocoeperative relationships with other microorganisms and with higher plants and animals. There is continuing interest in bioprospecting in all areas of the world, and major companies have been organized to continue to explore microbial diversity and identify microorganisms with new capabilities. **Uncultured microorganisms and microbial diversity (section 6.5)**

Genetic Manipulation of Microorganisms

Genetic manipulations are used to produce microorganisms with new and desirable characteristics. The classical methods of microbial genetics (*see chapter 13*) play a vital role in the development of cultures for industrial microbiology.

Mutation

Once a promising culture is found, a variety of techniques can be used for culture improvement, including chemical mutagens and ultraviolet light (*see chapter 11*). As an example, the first cultures of *Penicillium notatum*, which could be grown only under static conditions, yielded low concentrations of penicillin. In 1943 a strain of *Penicillium chrysogenum* was isolated—

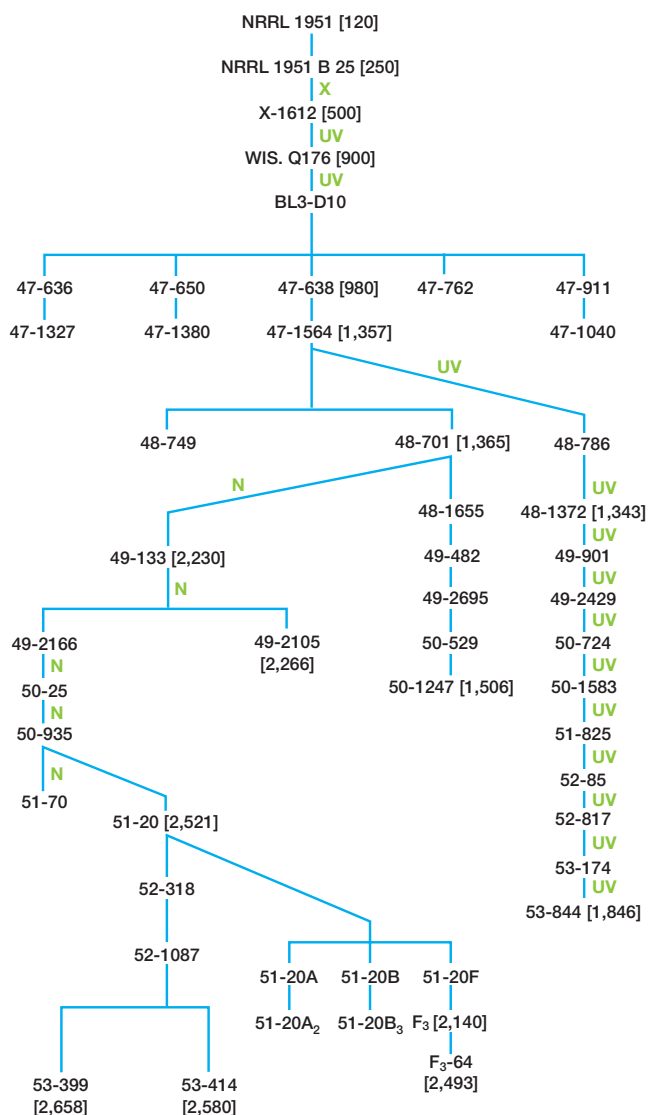


Figure 42.1 Mutation Makes It Possible to Increase Fermentation Yields. A “genealogy” of the mutation processes used to increase penicillin yields with *Penicillium chrysogenum* using X-ray treatment (X), UV treatment (UV), and mustard gas (N). By use of these mutational processes, the yield was increased from 120 International Units (IU) to 2,580 IU, a 20-fold increase. Unmarked transfers were used for mutant growth and isolation. Yields in international units/ml in brackets.

strain NRRL 1951—which was further improved through mutation (**figure 42.1**). Today most penicillin is produced with *Penicillium chrysogenum*, grown in aerobic stirred fermenters, which gives 55-fold higher penicillin yields than the original static cultures.

Protoplast Fusion

Protoplast fusion is now widely used with yeasts and molds. Most of these microorganisms are asexual or of a single mating type, which decreases the chance of random mutations that could lead to strain degeneration. To carry out genetic studies with these microorganisms, protoplasts are prepared by growing the cells in an isotonic solution while treating them with enzymes, including cellulase and beta-galacturonidase. The protoplasts are then regenerated using osmotic stabilizers such as sucrose. If fusion occurs to form hybrids, desired recombinants are identified by means of selective plating techniques. After regeneration of the cell wall, the new protoplasm fusion product can be used in further studies.

A major advantage of the protoplast fusion technique is that protoplasts of different microbial species can be fused, even if they are not closely linked taxonomically. For example, protoplasts of *Penicillium roquefortii* have been fused with those of *P. chrysogenum*. Even yeast protoplasts and erythrocytes can be fused.

Insertion of Short DNA Sequences

Short lengths of chemically synthesized DNA sequences can be inserted into recipient microorganisms by the process of **site-directed mutagenesis**. This can create small genetic alterations leading to a change of one or several amino acids in the target protein. Such minor amino acid changes have been found to lead, in many cases, to unexpected changes in protein characteristics, and have resulted in new products such as more environmentally resistant enzymes and enzymes that can catalyze desired reactions. These approaches are part of the field of **protein engineering**. [Site-directed mutagenesis \(p. 323\)](#)

Enzymes and bioactive peptides with markedly different characteristics (stability, kinetics, activities) can be created. The molecular basis for the functioning of these modified products also can be better understood. One of the most interesting areas is the design of enzyme-active sites to promote the modification of “unnatural substrates.” This approach may lead to improved transformation of recalcitrant materials, or even the degradation of materials that have previously not been amenable to biological processing.

1. How are industrial microbiology and biotechnology similar and different?
2. Based on recent estimates, what portion of the microorganisms have been cultured from soils and from aquatic and marine environments?
3. What is protoplast fusion and what types of microorganisms are used in this process?
4. Describe site-directed mutagenesis and how it is used in biotechnology.
5. What is protein engineering?

Table 42.3 Combinatorial Biology in Biotechnology: The Expression of Genes in Other Organisms to Improve Processes and Products

Property or Product Transferred	Microorganism Used	Combinatorial Process
Ethanol production 1,3-Propanediol production	<i>Escherichia coli</i> <i>E. coli</i>	Integration of pyruvate decarboxylase and alcohol dehydrogenase II from <i>Zymomonas mobilis</i> . Introduction of genes from the <i>Klebsiella pneumoniae</i> dha region into <i>E. coli</i> made possible anaerobic 1,3-propanediol production.
Cephalosporin precursor synthesis	<i>Penicillium chrysogenum</i>	Production 7-ADC and 7-ADCA ^a precursors by incorporation of the expandase gene of <i>Cephalosporin acremonium</i> into <i>Penicillium</i> by transformation.
Lactic acid production Xylitol production	<i>Saccharomyces cerevisiae</i> <i>S. cerevisiae</i>	A muscle bovine lactate dehydrogenase gene (LDH-A) expressed in <i>S. cerevisiae</i> . 95% xylitol conversion from xylose was obtained by transforming the XYLI gene of <i>Pichia stipitis</i> encoding a xylose reductase into <i>S. cerevisiae</i> , making this organism an efficient organism for the production of xylitol, which serves as a sweetener in the food industry.
Creatininase ^b	<i>E. coli</i>	Expression of the creatininase gene from <i>Pseudomonas putida</i> R565. Gene inserted with a pUC18 vector.
Pediocin ^c	<i>S. cerevisiae</i>	Expression of bacteriocin from <i>Pediococcus acidilactici</i> in <i>S. cerevisiae</i> to inhibit wine contaminants.
Acetone and butanol production	<i>Clostridium acetobutylicum</i>	Introduction of a shuttle vector into <i>C. acetobutylicum</i> by an improved electrotransformation protocol, which resulted in acetone and butanol formation.

^a7-ACA = 7-aminoccephalosporanic acid; 7-ADCA = 7-aminodeacetoxycephalosporonic acid.

^bT.-Y. Tang; C.-J. Wen; and W.-H. Liu. 2000. Expression of the creatininase gene from *Pseudomonas putida* RS65 in *Escherichia coli*. *J. Ind. Microbiol. Biotechnol.* 24:2–6.

^cH. Schoeman; M. A. Vivier; M. DuToit; L. M. Y. Dicks; and I. S. Pretorius. 1999. The development of bacteriocidal yeast strains by expressing the *Pediococcus acidilactici* pediocin gene (pedA) in *Saccharomyces cerevisiae*. *Yeast* 15:647–656.

Adapted from S. Ostergaard; L. Olsson; and J. Nielson. 2000. Metabolic engineering of *Saccharomyces cerevisiae*. *Microbiol. Mol. Biol. Rev.* 64(1):34–50.

Transfer of Genetic Information between Different Organisms

New alternatives have arisen through the transfer of nucleic acids between different organisms, which is part of the rapidly developing field of **combinatorial biology** (table 42.3). This involves the transfer of genes for the synthesis of a specific product from one organism into another, giving the recipient varied capabilities such as an increased capacity to carry out hydrocarbon degradation. An important early example of this approach was the creation of the “superbug,” patented by A. M. Chakrabarty in 1974, which had an increased capability of hydrocarbon degradation. The genes for antibiotic production can be transferred to a microorganism that produces another antibiotic, or even to a non-antibiotic-producing microorganism. For example, the genes for synthesis of bialaphos (an antibiotic herbicide) were transferred from *Streptomyces hygroscopicus* to *S. lividans*. Other examples are the expression in *E. coli*, of the enzyme creatininase from *Pseudomonas putida* and the production of pediocin, a bacteriocin, in a yeast used in wine fermentation for the purpose of controlling bacterial contaminants. **Bacteriocins** (pp. 297, 712)

DNA expression in different organisms can improve production efficiency and minimize the purification steps required before the product is ready for use. For example, recombinant baculoviruses (see p. 415) can be replicated in insect larvae to achieve rapid large-scale production of a desired virus or protein. Transgenic plants (discussed on pp. 335–36) may be used to manufacture large quantities of a variety of metabolic products. A most imaginative way of incorporating new DNA into a plant is to simply shoot it in using DNA-coated microprojectiles and a gene gun (see section 14.6).

A wide range of genetic information also can be inserted into microorganisms using vectors and recombinant DNA techniques. Vectors (see section 14.5) include artificial chromosomes such as those for yeasts (YACs), bacteria (BACs), P1 bacteriophage-derived chromosomes (PACs), and mammalian artificial chromosomes (MACs). YACs are especially valuable because large DNA sequences (over 100 kb) can be maintained in the YAC as a separate chromosome in yeast cells. A good example of vector use is provided by the virus that causes foot-and-mouth disease of cattle and other livestock. Genetic information for a foot-and-mouth disease virus antigen can be incorporated into *E. coli*, followed by the expression of this genetic information and synthesis of the gene product for use in vaccine production (figure 42.2).

Genetic information transfer allows the production of specific proteins and peptides without contamination by similar products that might be synthesized in the original organism. This approach can decrease the time and cost of recovering and purifying a product. Another major advantage of peptide production with modern biotechnology is that only biologically active stereoisomers are produced. This specificity is required to avoid the possible harmful side effects of inactive stereoisomers, as occurred in the thalidomide disaster.

In summary, modern gene-cloning techniques provide a considerable range of possibilities for manipulation of microorganisms and the use of plants and animals (or their cells) as factories for the expression of recombinant DNA. Newer molecular techniques continue to be discovered and applied to transfer genetic information and to construct microorganisms with new capabilities.

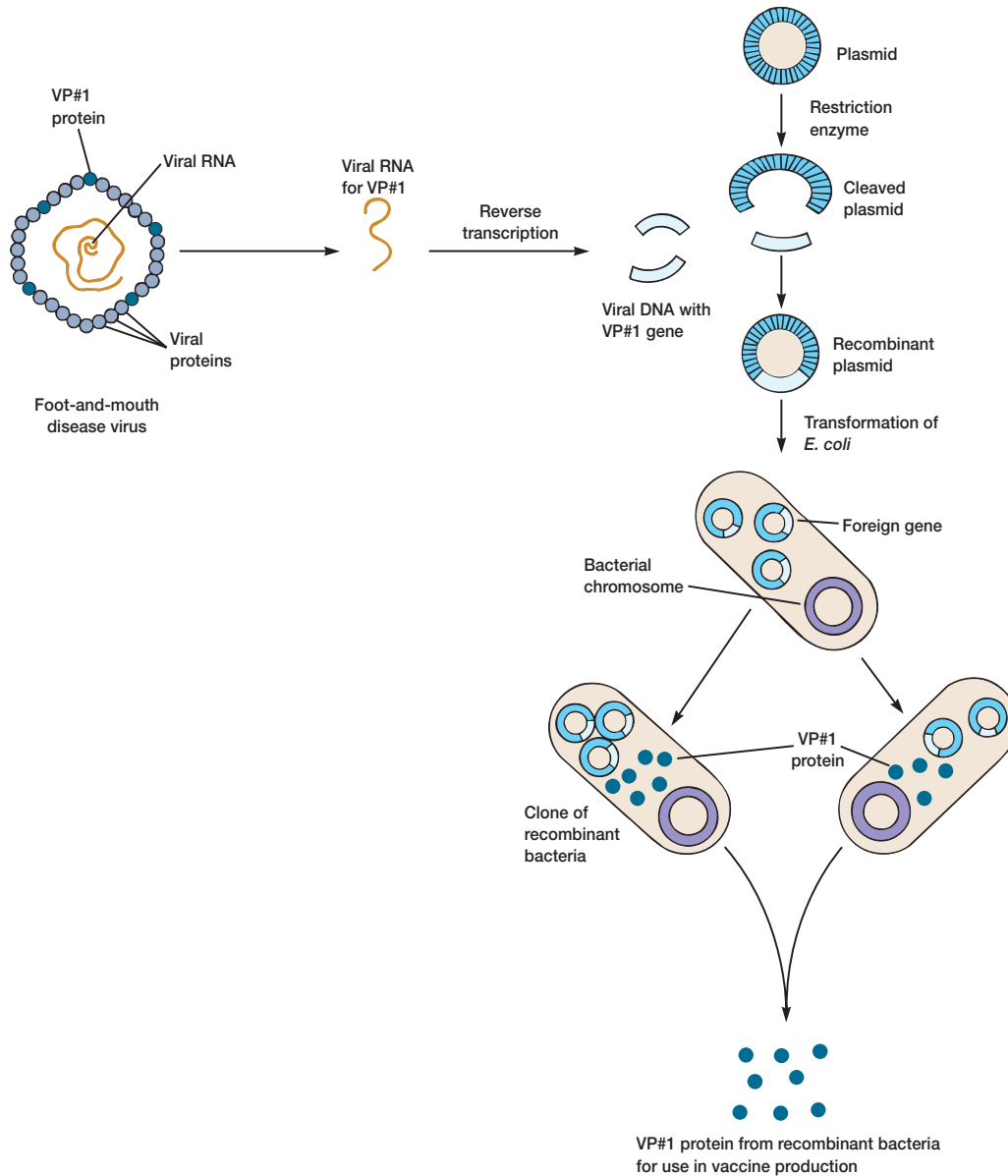


Figure 42.2 Recombinant Vaccine Production. Genes coding for desired products can be expressed in different organisms. By the use of recombinant DNA techniques, a foot-and-mouth disease vaccine is produced through cloning the vaccine genes into *Escherichia coli*.

Table 42.4 Examples of Recombinant DNA Systems Used to Modify Gene Expression

Product	Microorganism	Change
Actinorhodin	<i>Streptomyces coelicolor</i>	Modification of gene transcription
Cellulase	<i>Clostridium</i> genes in <i>Bacillus</i>	Amplification of secretion through chromosomal DNA amplification
Recombinant protein albumin	<i>Saccharomyces cerevisiae</i>	Fusion to a high-production protein
Heterologous protein	<i>Saccharomyces cerevisiae</i>	Use of the inducible strong hybrid promoter UAS _{gal} /CYC1
Enhanced growth rate ^a	<i>Aspergillus nidulans</i>	Overproduction of glyceraldehyde-3-phosphate dehydrogenase
Amino acids ^b	<i>Corynebacterium</i>	Isolation of biosynthetic genes that lead to enhanced enzyme activities or removal of feedback regulation

^{a,b}S. Ostergaard; L. Olsson; and J. Nielson. 2000. Metabolic engineering of *Saccharomyces cerevisiae*. *Microbiol. Mol. Biol. Rev.* 64(1):34–50. Table 1, p. 35

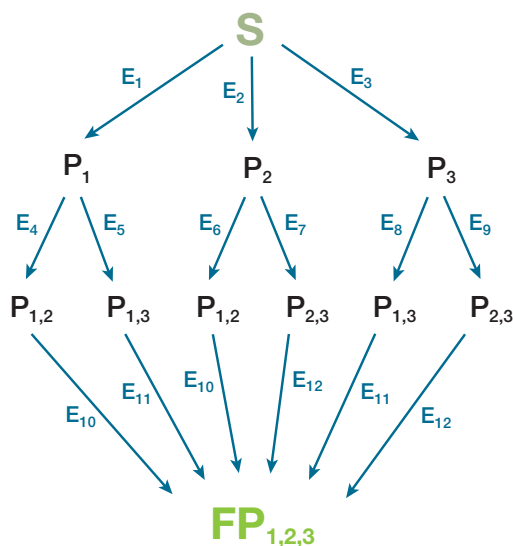


Figure 42.3 Pathway Architecture, a Critical Factor in Metabolic Engineering. Alternative steps for addition of three functional groups to a basic chemical skeleton may have different efficiencies, and it is critical to choose the most efficient combination of enzymatic steps or pathway to yield the desired product. E1 → E12 = different enzymes; P = intermediary products after the addition of the first and second functional groups, and FP = final product.

Modification of Gene Expression

In addition to inserting new genes in organisms, it also is possible to modify gene regulation by changing gene transcription, fusing proteins, creating hybrid promoters, and removing feedback regulation controls. These approaches make it possible to overproduce a wide variety of products, as shown in **table 42.4**. As a further example, genes for the synthesis of the antibiotic actinorhodin have been transferred into strains producing another antibiotic, resulting in the production of two antibiotics by the same cell.

This approach of modifying gene expression also can be used to intentionally alter metabolic pathways by inactivation or deregulation of specific genes, which is the field of **pathway architecture**, as shown in **figure 42.3**. Alternative routes can be used to add three functional groups to a molecule. Some of these pathways may be more efficient than the others. Understanding pathway architecture makes it possible to design a pathway that will be most efficient by avoiding slower or energetically more costly routes. This approach has been used to improve penicillin production by **metabolic pathway engineering (MPE)**.

An interesting recent development in modifying gene expression, which illustrates **metabolic control engineering**, is that of altering controls for the synthesis of lycopene, an important antioxidant normally present at high levels in tomatoes and tomato products. In this case, an engineered regulatory circuit was designed to control lycopene synthesis in response to the internal metabolic state of *E. coli*. An artificially engineered region that controls two key lycopene synthesis enzymes is stimulated by excess glycolytic activity and influences acetyl phosphate levels, thus allowing a significant increase in lycopene production while reducing negative impacts of metabolic imbalances.

Another recent development is the use of modified gene expression to produce variants of the antibiotic erythromycin. Blocking specific biochemical steps (**figure 42.4**) in pathways for the synthesis of an antibiotic precursor resulted in modified final products. These altered products, which have slightly different structures, can be tested for their possible antimicrobial effects. In addition, by the use of this approach, it is possible to better establish the structure-function relationships of antibiotics. Procedures for using microorganisms in the production of chemical feedstocks also have been developed using this MPE approach. By turning on and off

1. What is combinatorial biology and what is the basic approach used in this technique?
2. What types of major products have been created using combinatorial biology?
3. Why might one want to insert a gene in a foreign cell and how is this done?
4. Why is it important to produce specific isomers of products for use in animal and human health?

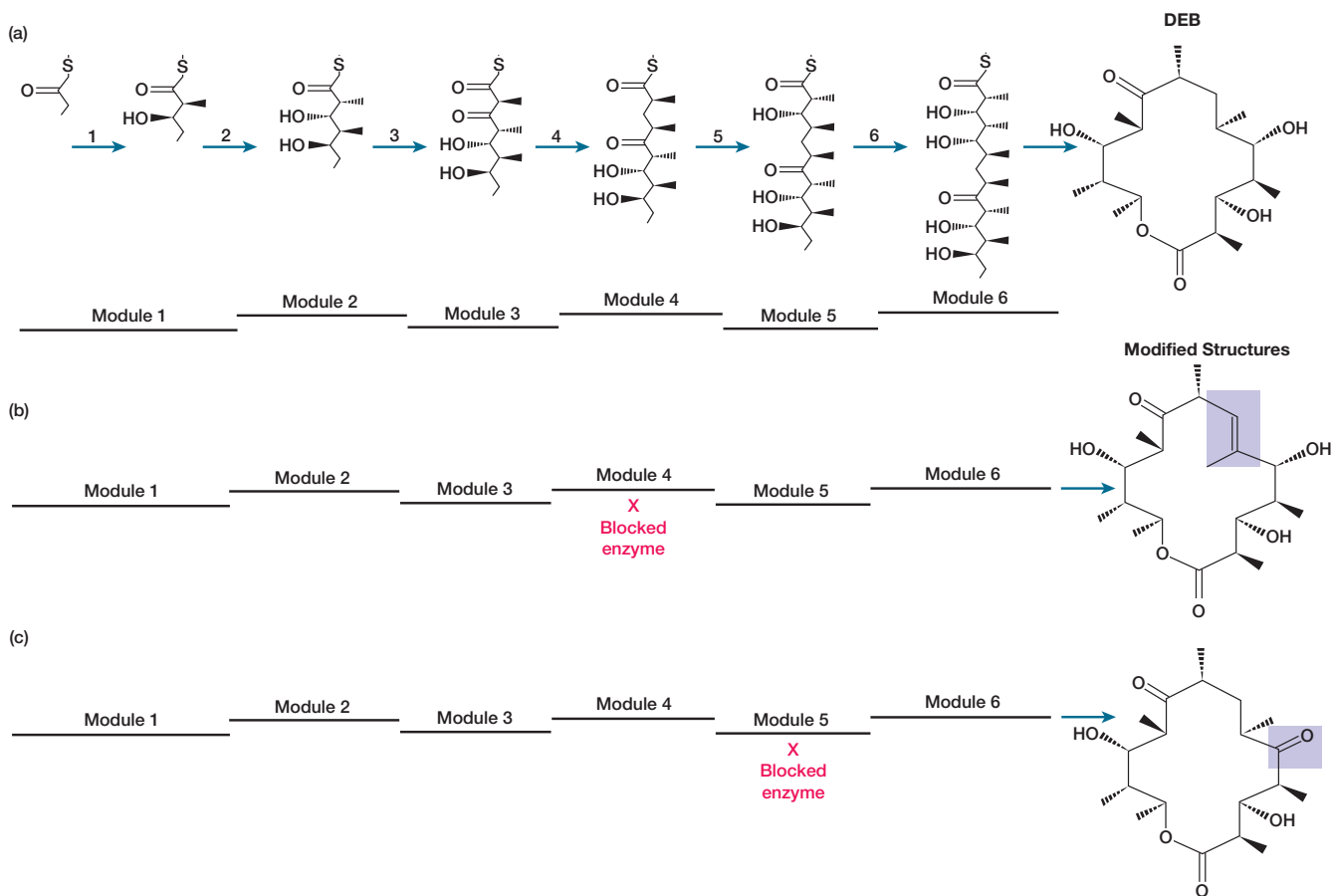


Figure 42.4 Metabolic Engineering to Create Modified Antibiotics. (a) Model for six elongation cycles (modules) in the normal synthesis of 6-deoxyerythronilide B (DEB), a precursor to the important antibiotic erythromycin. (b) Changes in structure that occur when the enoyl reductase enzyme of module 4 is blocked. (c) Changes in structure that occur when the keto reductase enzyme of module 5 is blocked. These changed structures (the highlighted areas) may lead to the synthesis of modified antibiotics with improved properties.

specific genes, feedstock chemicals such as 1,2-propanediol and 1,3-propanediol can be produced at high levels (**figure 42.5**). These particular chemicals are used in semimoist dog foods!

Other examples include the increased synthesis of antibiotics and cellulases, modification of gene expression, DNA amplification, greater protein synthesis, and interactive enzyme overproduction or removal of feedback inhibition. Recombinant plasminogen, for example, may comprise 20 to 40% of the soluble protein in a modified strain, a tenfold increase in concentration over that in the original strain.

Natural Genetic Engineering

The newest approach for creating new metabolic capabilities in a given microorganism is the area of **natural genetic engineering**, which employs **forced evolution** and **adaptive mutations**

(see p. 246). This is the process of using specific environmental stresses to “force” microorganisms to mutate and adapt, thus creating microorganisms with new biological capabilities. The mechanisms of these adaptive mutational processes include DNA rearrangements in which transposable elements and various types of recombination play critical roles, as shown in **table 42.5**.

Studies on natural genetic engineering are in a state of flux. It may be that “forced processes of evolution” are more effective than rational design in some cases. Such “environmentally directed” mutations have the potential of producing microbes with new degradative or biosynthetic capabilities.

Although there is much controversy concerning this area, the responses of microorganisms to stress provide the potential of generating microorganisms with new microbial capabilities for use in industrial microbiology and biotechnology.

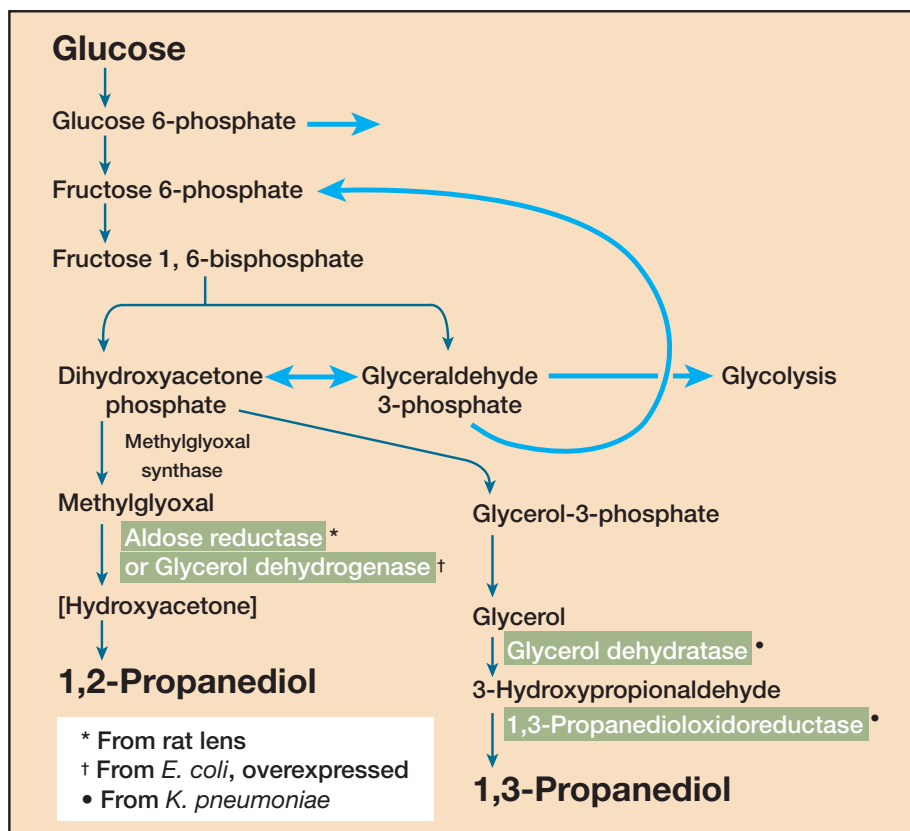


Figure 42.5 Use of Combinatorial Biology to Produce Propanediol in *E. coli*. Either an aldose reductase enzyme from rat lens or an overexpressed *E. coli* glycerol dehydrogenase enzyme and two enzymes from *Klebsiella pneumoniae*, glycerol dehydrogenase and 1,3-propanedioloxidoreductase (all green), are used to shift the intermediary metabolism of *E. coli* to the production of propanediols.

Table 42.5 Natural Genetic Engineering Systems in Bacteria

Genetic Engineering Mechanisms	DNA Changes Mediated
Localized SOS mutagenesis	Base substitutions, frameshifts
Adapted frameshifting	–1 frameshifting
Tn5, Tn9, Tn10 precise excision	Reciprocal recombination of flanking 8/9 bp repeats; restores original sequence
In vivo deletion, inversion, fusion, and duplication formation	Generally reciprocal recombination of short sequence repeats; occasionally nonhomologous
Type II topoisomerase recombination	Deletions and fusions by nonhomologous recombination, sometimes at short repeats
Site-specific recombination (type I topoisomerases)	Insertions, excisions/deletions, inversions by concerted or successive cleavage-ligation reactions at short sequence repeats; tolerates mismatches
Transposable elements (many species)	Insertions, transpositions, replicon fusions, adjacent deletions/excisions, adjacent inversions by ligation of 3' OH transposon ends of 5' PO ₄ groups from staggered cuts at nonhomologous target sites
DNA uptake (transformation competence)	Uptake of single strand independent of sequence, or of double-stranded DNA carrying species identifier sequence

Adapted from J. A. Shapiro, 1999. Natural genetic engineering, adaptive mutation, and bacterial evolution. In *microbial ecology of infectious disease*, E. Rosenberg, editor, 259–75. Washington, D.C.: American Society for Microbiology. Derived from Table 2, pp. 263–64.

Preservation of Microorganisms

Once a microorganism or virus has been selected or created to serve a specific purpose, it must be preserved in its original form for further use and study. Periodic transfers of cultures have been used in the past, although this can lead to mutations and pheno-

typic changes in microorganisms. To avoid these problems, a variety of culture preservation techniques may be used to maintain desired culture characteristics (table 42.6). **Lyophilization**, or freeze-drying, and storage in liquid nitrogen are frequently employed with microorganisms. Although lyophilization and liquid

Table 42.6 Methods Used to Preserve Cultures of Interest for Industrial Microbiology and Biotechnology

Method	Comments
Periodic transfer	Variables of periodic transfer to new media include transfer frequency, medium used, and holding temperature; this can lead to increased mutation rates and production of variants
Mineral oil slant	A stock culture is grown on a slant and covered with sterilized mineral oil; the slant can be stored at refrigerator temperature
Minimal medium, distilled water, or water agar	Washed cultures are stored under refrigeration; these cultures can be viable for 3 to 5 months or longer
Freezing in growth media	Not reliable; can result in damage to microbial structures; with some microorganisms, however, this can be a useful means of culture maintenance
Drying	Cultures are dried on sterile soil (soil stocks), on sterile filter paper disks, or in gelatin drops; these can be stored in a desiccator at refrigeration temperature, or frozen to improve viability
Freeze-drying (lyophilization)	Water is removed by sublimation, in the presence of a cryoprotective agent; sealing in an ampule can lead to long-term viability, with 30 years having been reported
Ultrafreezing	Liquid nitrogen at -196°C is used, and cultures of fastidious microorganisms have been preserved for more than 15 years

nitrogen storage are complicated and require expensive equipment, they do allow microbial cultures to be stored for years without loss of viability or an accumulation of mutations.

1. What types of recombinant DNA techniques are being used to modify gene expression in microorganisms?
2. Define metabolic control engineering, metabolic pathway engineering, forced evolution, and adaptive mutations.
3. Why might natural genetic engineering be useful in modern microbial biotechnology?
4. What approaches can be used for the preservation of microorganisms?

42.2 Microorganism Growth in Controlled Environments

For many industrial processes, microorganisms must be grown using specifically designed media under carefully controlled conditions, including temperature, aeration, and nutrient feeding during the course of the fermentation. The growth of microorganisms under such controlled environments is expensive, and this approach is used only when the desired product can be sold for a profit. These high costs arise from the expense of development of the particular microorganism to be used in a large-scale fermentation, the equipment, medium preparation, product purification and packaging, and marketing efforts. In addition, if this is a product to be used in animal or human health care, literally millions of dollars must be spent conducting trials and obtaining regulatory approval before even a dollar of income is available to investors. Patents are obtained whenever possible to assure that investment costs can be recovered over a longer time period. Clearly products that are brought to market must have a high monetary value. The development of appropriate culture media and the growth of microorganisms under industrial conditions are the subjects of this section.

Table 42.7 Fermentation: A Word with Many Meanings for the Microbiologist

1. Any process involving the mass culture of microorganisms, either aerobic or anaerobic
2. Any biological process that occurs in the absence of O_2
3. Food spoilage
4. The production of alcoholic beverages
5. Use of an organic substrate as the electron donor and acceptor
6. Use of an organic substrate as a reductant, and of the same partially degraded organic substrate as an oxidant
7. Growth dependent on substrate-level phosphorylation

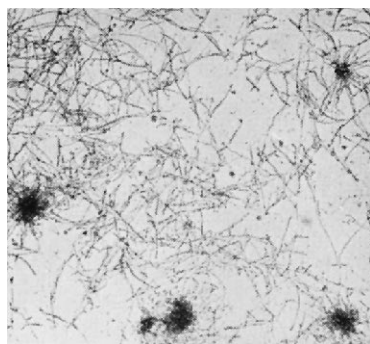
Before proceeding, it is necessary to clarify terminology. The term **fermentation**, used in a physiological sense in earlier sections of the book, is employed in a much more general way in relation to industrial microbiology and biotechnology. As noted in **table 42.7**, the term can have several meanings, including the mass culture of microorganisms (or even plant and animal cells). The development of industrial fermentations requires appropriate culture media and the large-scale screening of microorganisms. Often years are needed to achieve optimum product yields. Many isolates are tested for their ability to synthesize a new product in the desired quantity. Few are successful. [Fermentation as a physiological process \(pp. 179–81\)](#)

Medium Development

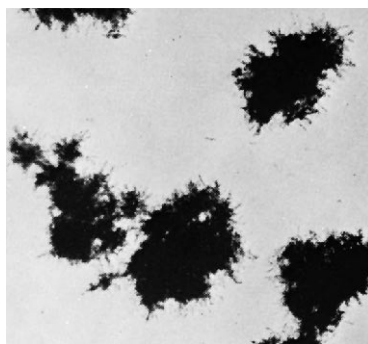
The medium used to grow a microorganism is critical because it can influence the economic competitiveness of a particular process. Frequently, lower-cost crude materials are used as sources of carbon, nitrogen, and phosphorus (**table 42.8**). Crude plant hydrolysates often are used as complex sources of carbon, nitrogen, and growth factors. By-products from the brewing industry frequently are employed because of their lower cost and greater availability. Other useful carbon sources include molasses and whey from cheese manufacture. [Microbial growth media \(pp. 104–6\)](#)

Table 42.8 Major Components of Growth Media Used in Industrial Processes

Source	Raw Material	Source	Raw Material
Carbon and energy	Molasses	Vitamins	Crude preparations of plant and animal products
	Whey		
	Grains		
	Agricultural wastes (corn cobs)		
Nitrogen	Corn-steep liquor	Iron, trace salts	Crude inorganic chemicals
	Soybean meal	Buffers	Chalk or crude carbonates
	Stick liquor (slaughterhouse products)		Fertilizer-grade phosphates
	Ammonia and ammonium salts	Antifoam agents	Higher alcohols
	Nitrates		Silicones
	Distiller's solubles		Natural esters
			Lard and vegetable oils



(a)



(b)

Figure 42.6 Filamentous Growth During Fermentation. Filamentous fungi and actinomycetes can change their growth form during the course of a fermentation. The development of pelleted growth by fungi has major effects on oxygen transfer and energy required to agitate the culture. (a) Initial culture. (b) after 18 hours growth.

The levels and balance of minerals (especially iron) and growth factors can be critical in medium formulation. For example, biotin and thiamine, by influencing biosynthetic reactions, control product accumulation in many fermentations. The medium also may be designed so that carbon, nitrogen, phosphorus, iron, or a specific growth factor will become limiting after a given time during the fermentation. In such cases the limitation often causes a shift from growth to production of desired metabolites.

Growth of Microorganisms in an Industrial Setting

Once a medium is developed, the physical environment for microbial functioning in the mass culture system must be defined. This often involves precise control of agitation, temperature, pH changes, and oxygenation. Phosphate buffers can be used to control pH while also functioning as a source of phosphorus. Oxygen limitations, especially, can be critical in aerobic growth processes. The O_2 concentration and flux rate must be sufficiently high to have O_2 in excess within the cells so that it is not limiting. This is especially true when a dense microbial culture is growing. When filamentous fungi and actinomycetes are cultured, aeration can be even further limited by filamentous growth (**figure 42.6**). Such filamentous growth results in a viscous, plastic medium, known as a **non-Newtonian broth**, which offers even more resistance to stir-

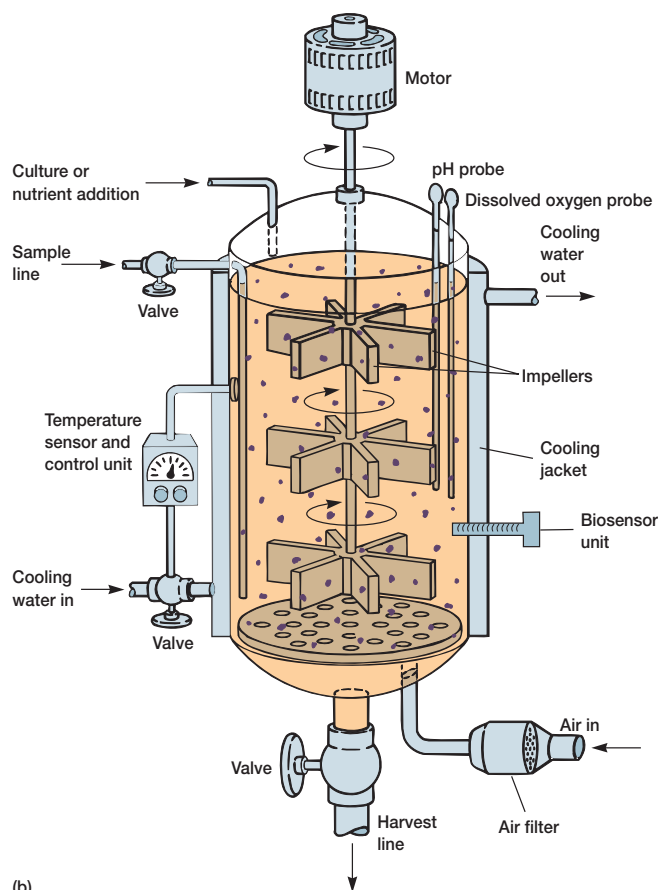
ring and aeration. To minimize this problem, cultures can be grown as pellets or flocs or bound to artificial particles.

It is essential to assure that these physical factors are not limiting microbial growth. This is most critical during **scaleup**, where a successful procedure developed in a small shake flask is modified for use in a large fermenter. One must understand the microenvironment of the culture and maintain similar conditions near the individual cell despite increases in the culture volume. If a successful transition can be made from a process originally developed in a 250 ml Erlenmeyer flask to a 100,000 liter reactor, then the process of scaleup has been carried out properly.

Microorganisms can be grown in culture tubes, shake flasks, and stirred fermenters or other mass culture systems. Stirred fermenters can range in size from 3 or 4 liters to 100,000 liters or larger, depending on production requirements (**figure 42.7**). A typical industrial stirred fermentation unit is illustrated in **figure 42.7b**. This unit requires a large capital investment and skilled operators. All required steps in the growth and harvesting of products must be carried out under aseptic conditions. Not only must the medium be sterilized but aeration, pH adjustment, sampling, and process monitoring must be carried out under rigorously controlled conditions. When required, foam control agents must be added, especially with high-protein media. Computers are commonly used to monitor outputs from probes that determine microbial biomass, levels of critical metabolic



(a)



(b)

Figure 42.7 Industrial Stirred Fermenters. (a) Large fermenters used by a pharmaceutical company for the microbial production of antibiotics. (b) Details of a fermenter unit. This unit can be run under aerobic or anaerobic conditions, and nutrient additions, sampling, and fermentation monitoring can be carried out under aseptic conditions. Biosensors and infrared monitoring can provide real-time information on the course of the fermentation. Specific substrates, metabolic intermediates, and final products can be detected.

products, pH, input and exhaust gas composition, and other parameters. Such information is needed for precise process and product control. Environmental conditions can be changed or held constant over time, depending on the goals for the particular process.

Frequently a critical component in the medium, often the carbon source, is added continuously—**continuous feed**—so that the microorganism will not have excess substrate available at any given time. An excess of substrate can cause undesirable metabolic waste products to accumulate. This is particularly important with glucose and other carbohydrates. If excess glucose is present at the beginning of a fermentation, it can be catabolized to yield ethanol, which is lost as a volatile product and reduces the final yield. This can occur even under aerobic conditions.

Besides the traditional stirred aerobic or anaerobic fermenter, other approaches can be used to grow microorganisms. These alternatives, illustrated in **figure 42.8**, include lift-tube fermenters (**figure 42.8a**), which eliminate the need for stirrers that can be fouled by filamentous fungi. Also available is solid-state fermentation (**figure 42.8b**), in which the substrate is not diluted in water. In various types of fixed- (**figure 42.8c**) and fluidized-bed reactors (**figure 42.8d**), the microorganisms are associated with inert surfaces as biofilms (*see pp. 620–22*), and medium flows past the fixed or suspended particles.

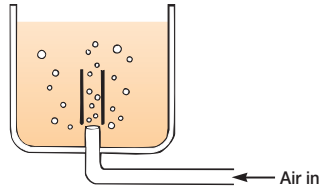
Dialysis culture units also can be used (**figure 42.8e**). These units allow toxic waste metabolites or end products to diffuse away from the microbial culture and permit new substrates to diffuse through the membrane toward the culture. Continuous culture techniques using chemostats (**figure 42.8f**) can markedly improve cell outputs and rates of substrate use because microorganisms can be maintained in a continuous logarithmic phase. However, continuous maintenance of an organism in an active growth phase is undesirable in many industrial processes.

Microbial products often are classified as primary and secondary metabolites. As shown in **figure 42.9**, **primary metabolites** consist of compounds related to the synthesis of microbial cells in the growth phase. They include amino acids, nucleotides, and fermentation end products such as ethanol and organic acids. In addition, industrially useful enzymes, either associated with the microbial cells or exoenzymes, often are synthesized by microorganisms during growth. These enzymes find many uses in food production and textile finishing.

Secondary metabolites usually accumulate during the period of nutrient limitation or waste product accumulation that follows the active growth phase. These compounds have no direct relationship to the synthesis of cell materials and normal growth. Most antibiotics and the mycotoxins fall into this category.

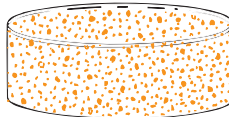
(a) Lift-tube fermenter

Density difference of
gas bubbles entrained
in medium results in
fluid circulation



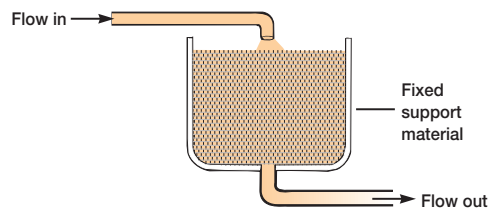
(b) Solid-state fermentation

Growth of culture
without presence of
added free water



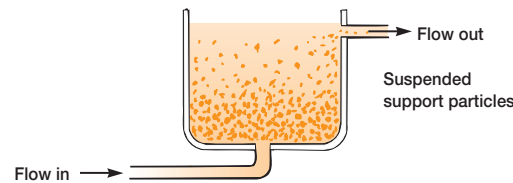
(c) Fixed-bed reactor

Microorganisms on surfaces
of support material;
flow can be up or down



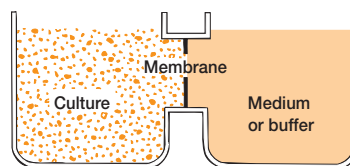
(d) Fluidized-bed reactor

Microorganisms on surfaces
of particles suspended
in liquid or gas stream—
upward flow



(e) Dialysis culture unit

Waste products diffuse
away from the culture.
Substrate may diffuse
through membrane to
the culture



(f) Continuous culture unit (Chemostat)

Medium in and excess
medium to waste with
wasted cells

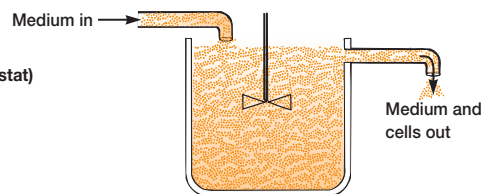


Figure 42.8 Alternate Methods for Mass Culture.
In addition to stirred fermenters, other methods can be
used to culture microorganisms in industrial processes.
In many cases these alternate approaches will have
lower operating costs and can provide specialized
growth conditions needed for product synthesis.

1. How is the cost of media reduced during industrial operations?
Discuss the effect of changing balances in nutrients such as minerals,
growth factors, and the sources of carbon, nitrogen, and phosphorus.
2. What factors increase the costs of microbial products, such as
antibiotics, used in animal and human health?
3. What are non-Newtonian broths, and why are these important in
fermentations?

4. Discuss scaleup and the objective of the scaleup process.
5. What parameters can be monitored in a modern, large-scale
industrial fermentation?
6. Besides the aerated, stirred fermenter, what other alternatives are
available for the mass culture of microorganisms in industrial
processes? What is the principle by which a dialysis culture
system functions?

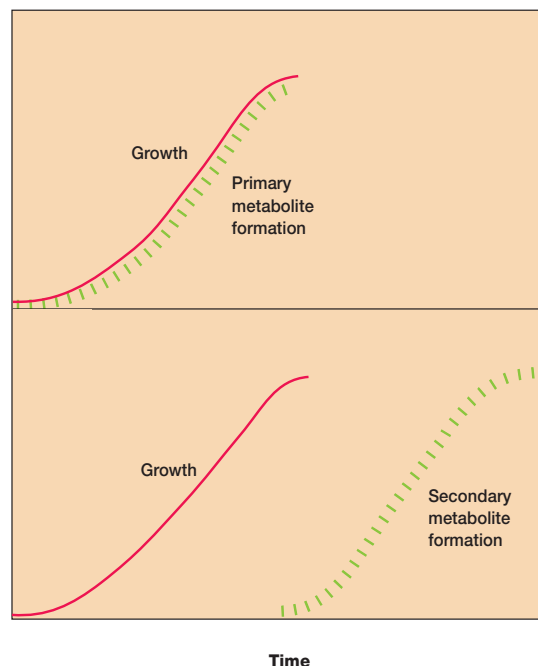


Figure 42.9 Primary and Secondary Metabolites. Depending on the particular organism, the desired product may be formed during or after growth. Primary metabolites are formed during the active growth phase, whereas secondary metabolites are formed after growth is completed.

42.3 Major Products of Industrial Microbiology

Industrial microbiology has provided products that have impacted our lives in many direct and often not appreciated ways. These products have profoundly changed our lives and life spans. They include industrial and agricultural products, food additives, medical products for human and animal health, and biofuels (**table 42.9**). Particularly, in the last few years, nonantibiotic compounds used in medicine and health have made major contributions to the improved well-being of animal and human populations. Only major products in each category will be discussed here.

Antibiotics

Many antibiotics are produced by microorganisms, predominantly by actinomycetes in the genus *Streptomyces* and by filamentous fungi (see **table 35.2**). In this chapter, the synthesis of several of the most important antibiotics will be discussed to illustrate the critical role of medium formulation and environmental control in the production of these important compounds. [Antibiotics in medicine](#) (**chapter 35**)

Penicillin

Penicillin, produced by *Penicillium chrysogenum*, is an excellent example of a fermentation for which careful adjustment of the medium composition is used to achieve maximum yields. Rapid production of cells, which can occur when high levels of glucose are used as a carbon source, does not lead to maximum antibiotic

Table 42.9 Major Microbial Products and Processes of Interest in Industrial Microbiology and Biotechnology

Substances	Microorganisms
Industrial Products	
Ethanol (from glucose)	<i>Saccharomyces cerevisiae</i>
Ethanol (from lactose)	<i>Kluyveromyces fragilis</i>
Acetone and butanol	<i>Clostridium acetobutylicum</i>
2,3-butanediol	<i>Enterobacter</i> , <i>Serratia</i>
Enzymes	<i>Aspergillus</i> , <i>Bacillus</i> , <i>Mucor</i> , <i>Trichoderma</i>
Agricultural Products	
Gibberellins	<i>Gibberella fujikuroi</i>
Food Additives	
Amino acids (e.g., lysine)	<i>Corynebacterium glutamicum</i>
Organic acids (citric acid)	<i>Aspergillus niger</i>
Nucleotides	<i>Corynebacterium glutamicum</i>
Vitamins	<i>Ashbya</i> , <i>Eremothecium</i> , <i>Blakeslea</i>
Polysaccharides	<i>Xanthomonas</i>
Medical Products	
Antibiotics	<i>Penicillium</i> , <i>Streptomyces</i> , <i>Bacillus</i>
Alkaloids	<i>Claviceps purpurea</i>
Steroid transformations	<i>Rhizopus</i> , <i>Arthrobacter</i>
Insulin, human growth hormone, somatostatin, interferons	<i>Escherichia coli</i> , <i>Saccharomyces cerevisiae</i> , and others (recombinant DNA technology)
Biofuels	
Hydrogen	Photosynthetic microorganisms
Methane	<i>Methanobacterium</i>
Ethanol	<i>Zymomonas</i> , <i>Thermoanaerobacter</i>

yields. Provision of the slowly hydrolyzed disaccharide lactose, in combination with limited nitrogen availability, stimulates a greater accumulation of penicillin after growth has stopped (figure 42.10). The same result can be achieved by using a slow continuous feed of glucose. If a particular penicillin is needed, the specific precursor

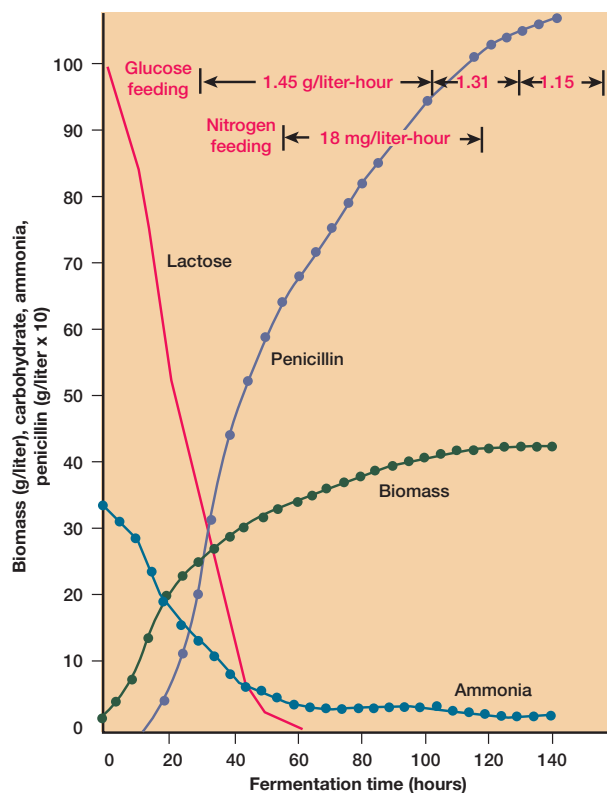


Figure 42.10 Penicillin Fermentation Involves Precise Control of Nutrients. The synthesis of penicillin begins when nitrogen from ammonia becomes limiting. After most of the lactose (a slowly catabolized disaccharide) has been degraded, glucose (a rapidly used monosaccharide) is added along with a low level of nitrogen. This stimulates maximum transformation of the carbon sources to penicillin.

is added to the medium. For example, phenylacetic acid is added to maximize production of penicillin G, which has a benzyl side chain (see figure 35.7). This “steering” process is used to maximize the production of desired compounds. The fermentation pH is maintained around neutrality by the addition of sterile alkali, which assures maximum stability of the newly synthesized penicillin. Once the fermentation is completed, normally in 6 to 7 days, the broth is separated from the fungal mycelium and processed by absorption, precipitation, and crystallization to yield the final product. This basic product can then be modified by chemical procedures to yield a variety of **semisynthetic penicillins**.

Streptomycin

Streptomycin is a secondary metabolite produced by *Streptomyces griseus*, for which changes in environmental conditions and substrate availability also influence final product accumulation. In this fermentation a soybean-based medium is used with glucose as a carbon source. The nitrogen source is thus in a combined form (soybean meal), which limits growth. After growth the antibiotic levels in the culture begin to increase (figure 42.11) under conditions of controlled nitrogen limitation.

The field of antibiotic development continues to expand. At present, 6,000 antibiotics have been described, with 4,000 of these derived from actinomycetes. About 300 new antibiotics are being discovered per year.

Amino Acids

Amino acids such as lysine and glutamic acid are used in the food industry as nutritional supplements in bread products and as flavor-enhancing compounds such as monosodium glutamate (MSG).

Amino acid production is typically carried out by means of **regulatory mutants**, which have a reduced ability to limit synthesis of an end product. The normal microorganism avoids overproduction of biochemical intermediates by the careful regulation of cellular metabolism. Production of glutamic acid and several other amino acids in large quantities is now carried out using mutants of

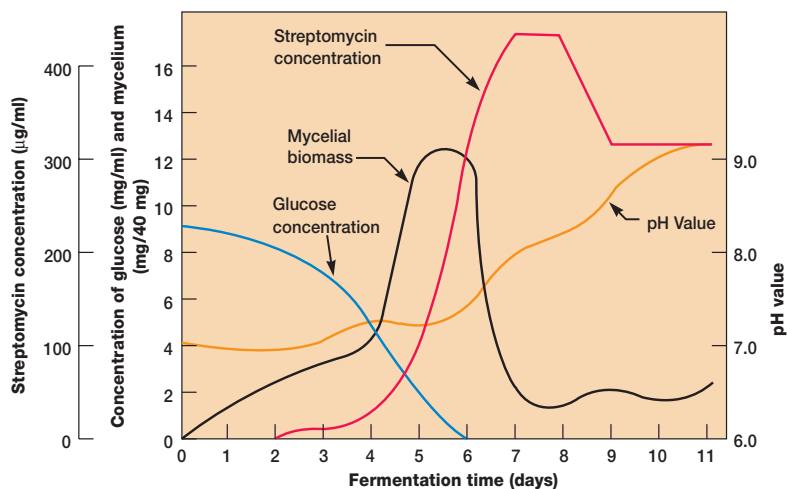


Figure 42.11 Streptomycin Production by *Streptomyces griseus*. Depletion of glucose leads to maximum antibiotic yields.

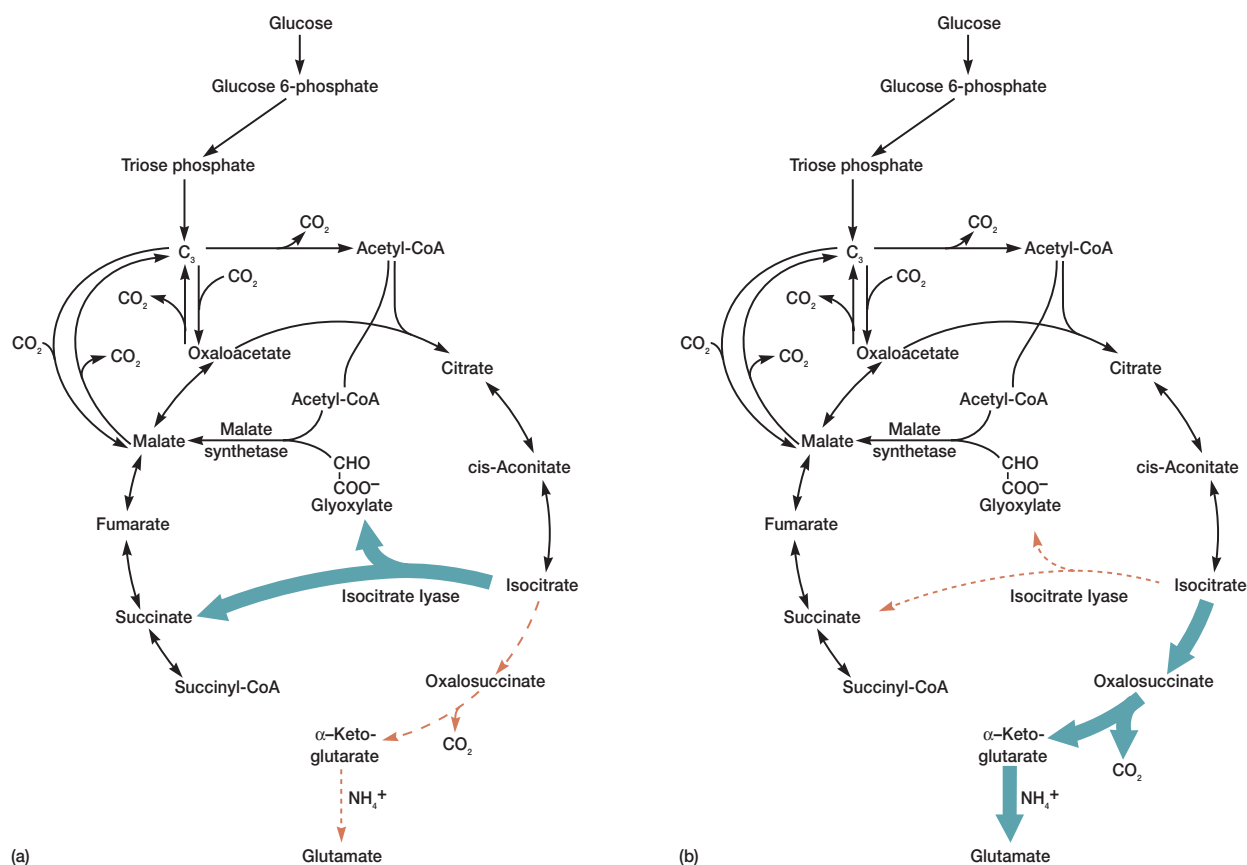


Figure 42.12 Glutamic Acid Production. The sequence of biosynthetic reactions leading from glucose to the accumulation of glutamate by *Corynebacterium glutamicum*. Major carbon flows are noted by bold arrows. (a) Growth with use of the glyoxylate bypass to provide critical intermediates in the TCA cycle. (b) After growth is completed, most of the substrate carbon is processed to glutamate (note shifted bold arrows). The dashed lines indicate reactions that are being used to a lesser extent.

Corynebacterium glutamicum that lack, or have only a limited ability to process, the TCA cycle intermediate α -ketoglutarate (see appendix II) to succinyl-CoA as shown in figure 42.12. A controlled low biotin level and the addition of fatty acid derivatives results in increased membrane permeability and excretion of high concentrations of glutamic acid. The impaired bacteria use the glyoxylate pathway (see section 10.6) to meet their needs for essential biochemical intermediates, especially during the growth phase. After growth becomes limited because of changed nutrient availability, an almost complete molar conversion (or 81.7% weight conversion) of isocitrate to glutamate occurs.

Lysine, an essential amino acid used to supplement cereals and breads, was originally produced in a two-step microbial process. This has been replaced by a single-step fermentation in which the bacterium *Corynebacterium glutamicum*, blocked in the synthesis of homoserine, accumulates lysine. Over 44 g/liter can be produced in a 3 day fermentation.

Although not used extensively in the United States, microorganisms with related regulatory mutations have been employed to produce a series of 5' purine nucleotides that serve as flavor enhancers for soups and meat products.

Organic Acids

Organic acid production by microorganisms is important in industrial microbiology and illustrates the effects of trace metal levels and balances on organic acid synthesis and excretion. Citric, acetic, lactic, fumaric, and gluconic acids are major products (table 42.10). Until microbial processes were developed, the major source of citric acid was citrus fruit from Italy. Today most citric acid is produced by microorganisms; 70% is used in the food and beverage industry, 20% in pharmaceuticals, and the balance in other industrial applications.

The essence of citric acid fermentation involves limiting the amounts of trace metals such as manganese and iron to stop As-

Table 42.10 Major Organic Acids Produced by Microbial Processes

Product	Microorganism Used	Representative Uses	Fermentation Conditions
Acetic acid	<i>Acetobacter</i> with ethanol solutions	Wide variety of food uses	Single-step oxidation, with 15% solutions produced; 95–99% yields
Citric acid	<i>Aspergillus niger</i> in molasses-based medium	Pharmaceuticals, as a food additive	High carbohydrate concentrations and controlled limitation of trace metals; 60–80% yields
Fumaric acid	<i>Rhizopus nigricans</i> in sugar-based medium	Resin manufacture, tanning, and sizing	Strongly aerobic fermentation; carbon-nitrogen ratio is critical; zinc should be limited; 60% yields
Gluconic acid	<i>Aspergillus niger</i> in glucose-mineral salts medium	A carrier for calcium and sodium	Uses agitation or stirred fermenters; 95% yields
Itaconic acid	<i>Aspergillus terreus</i> in molasses-salts medium	Esters can be polymerized to make plastics	Highly aerobic medium, below pH 2.2; 85% yields
Kojic acid	<i>Aspergillus flavus-oryzae</i> in carbohydrate-inorganic N medium	The manufacture of fungicides and insecticides when complexed with metals	Iron must be carefully controlled to avoid reaction with kojic acid after fermentation
Lactic acid	Homofermentative <i>Lactobacillus delbrueckii</i>	As a carrier for calcium and as an acidifier	Purified medium used to facilitate extraction

pergillus niger growth at a specific point in the fermentation. The medium often is treated with ion exchange resins to ensure low and controlled concentrations of available metals. Citric acid fermentation, which earlier was carried out by means of static surface growth, now takes place in aerobic stirred fermenters. Generally, high sugar concentrations (15 to 18%) are used, and copper has been found to counteract the inhibition of citric acid production by iron above 0.2 ppm. The success of this fermentation depends on the regulation and functioning of the glycolytic pathway and the tricarboxylic acid cycle (see section 9.4). After the active growth phase, when the substrate level is high, citrate synthase activity increases and the activities of aconitase and isocitrate dehydrogenase decrease. This results in citric acid accumulation and excretion by the stressed microorganism.

In comparison, the production of gluconic acid involves a single microbial enzyme, glucose oxidase, found in *Aspergillus niger*. *A. niger* is grown under optimum conditions in a corn-steep liquor medium. Growth becomes limited by nitrogen, and the resting cells transform the remaining glucose to gluconic acid in a single-step reaction. Gluconic acid is used as a carrier for calcium and iron and as a component of detergents.

Specialty Compounds for Use in Medicine and Health

In addition to the bulk products that have been produced over the last 30 to 40 years, such as antibiotics, amino acids, and organic acids, microorganisms are used for the production of nonantibiotic specialty compounds. These include sex hormones, antitumor agents, ionophores, and special compounds that influence bacteria, fungi, amoebae, insects, and plants (table 42.11). In all cases, it is necessary to produce and recover the products under carefully controlled conditions to assure that these medically important compounds reach the consumer in a stable, effective condition.

1. Approximately how many new antibiotics are being discovered per year? What portion of these are derived from actinomycetes?
2. What is the principal limitation created to stimulate citric acid accumulation by *Aspergillus niger*?
3. What types of nutrient limitations are often used in carrying out a successful fermentation? Consider carbon and nitrogen sources.
4. What critical limiting factors are used in the penicillin and streptomycin fermentations?
5. Give some important specialty compounds that are produced by the use of microorganisms.

Biopolymers

Biopolymers are microbially produced polymers used to modify the flow characteristics of liquids and to serve as gelling agents. These are employed in many areas of the pharmaceutical and food industries. The advantage of using microbial biopolymers is that production is independent of climate, political events that can limit raw material supplies, and the depletion of natural resources. Production facilities also can be located near sources of inexpensive substrates (e.g., near agricultural areas). [Bacterial exopolysaccharides \(p. 61\)](#)

At least 75% of all polysaccharides are used as stabilizers, for the dispersion of particulates, as film-forming agents, or to promote water retention in various products. Polysaccharides help maintain the texture of many frozen foods, such as ice cream, that are subject to drastic temperature changes. These polysaccharides must maintain their properties under the pH conditions in the particular food and be compatible with other polysaccharides. They should not lose their physical characteristics if heated.

Biopolymers include (1) dextrans, which are used as blood expanders and absorbents; (2) *Erwinia* polysaccharides that are in

Table 42.11 Nonantibiotic Specialty Compounds Produced by Microorganisms

Compound Type	Source	Specific Product	Process/Organism Affected
Polyethers	<i>Streptomyces cinnamonensis</i>	Monensin	Coccidiostat, rumenal growth promoter
	<i>S. lasaliensis</i>	Lasalocid	Coccidiostat, ruminal growth promoter
	<i>S. albus</i>	Salinomycin	Coccidiostat, ruminal growth promoter
Avermectins	<i>S. avermitilis</i>		Helminths and arthropods
Statins	<i>Aspergillus terreus</i>	Lovastatin	Cholesterol-lowering agent
	<i>Penicillium citrinum</i> + actinomycete ^a	Pravastatin	Cholesterol-lowering agent
Enzyme inhibitors	<i>S. clavuligerus</i>	Clavulanic acid	Penicillinase inhibitor
	<i>Actinoplanes</i> sp.	Acarbose	Intestinal glucosidase inhibitor (decreases hyperglycemia and triglyceride synthesis)
Bioherbicide	<i>S. hygroscopicus</i>	Bialaphos	
Immunosuppressants	<i>Tolypocladium inflatum</i>	Cyclosporin A	Organ transplants
	<i>S. tsukabensis</i>	FK-506	Organ transplants
	<i>S. hygroscopicus</i>	Rapamycin	Organ transplants
Anabolic agents	<i>Gibberella zeae</i>	Zearalenone	Farm animal medication
Uterocontractants	<i>Claviceps purpurea</i>	Ergot alkaloids	Induction of labor
Antitumor agents	<i>S. peuceticus</i> subsp. <i>caesius</i>	Doxorubicin	Cancer treatment
	<i>S. peuceticus</i>	Daunorubicin	Cancer treatment
	<i>S. caespitosus</i>	Mitomycin	Cancer treatment
	<i>S. verticillus</i>	Bleomycin	Cancer treatment

^aCompactin, produced by *Penicillium citrinum*, is changed to pravastatin by an actinomycete bioconversion.

Based on: A. L. Demain. 2000. Microbial biotechnology. *Tibtech* 18:26–31; A. L. Demain. 2000. Pharmaceutically active secondary metabolites of microorganisms. *App. Microbiol. Biotechnol.* 52:455–463; G. Lancini; A. L. Demain. 1999. Secondary metabolism in bacteria: Antibiotic pathways regulation, and function. In *Biology of the prokaryotes*, J. W. Lengeler, G. Drews, and H. G. Schlegel, editors, 627–51. New York: Thieme.

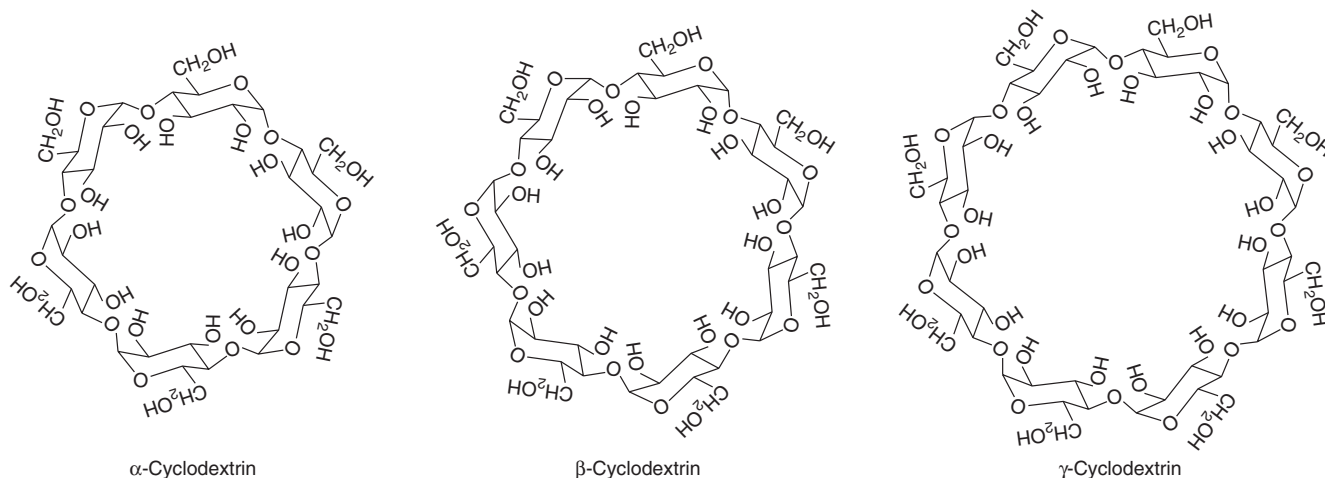


Figure 42.13 Cyclodextrins. The basic structures of cyclodextrins produced by *Thermoanaerobacter* are illustrated here. These unique oligopolysaccharides have many applications in medicine and industry.

paints; and (3) polyesters, derived from *Pseudomonas oleovorans*, which are a feedstock for specialty plastics. Cellulose microfibrils, produced by an *Acetobacter* strain, are used as a food thickener. Polysaccharides such as scleroglucan are used by the oil industry as drilling mud additives. Xanthan polymers enhance oil recovery by improving water flooding and the displacement of oil. This use

of xanthan gum, produced by *Xanthomonas campestris*, represents a large potential market for this microbial product.

The cyclodextrins have a unique structure, as shown in **figure 42.13**. They are cyclic oligosaccharides whose sugars are joined by α -1,4 linkages. Cyclodextrins can be used for a wide variety of purposes because these cyclical molecules bind with

substances and modify their physical properties. For example, cyclodextrins will increase the solubility of pharmaceuticals, reduce their bitterness, and mask chemical odors. Cyclodextrins also can be used as selective adsorbents to remove cholesterol from eggs and butter or protect spices from oxidation.

Biosurfactants

Many surfactants that have been used for commercial purposes are products of synthetic chemistry. At the present time there is an increasing interest in the use of biosurfactants. These are especially important for environmental applications where biodegradability is a major requirement. Biosurfactants are used for emulsification, increasing detergency, wetting and phase dispersion, as well as for solubilization. These properties are especially important in bioremediation, oil spill dispersion, and enhanced oil recovery (EOR).

The most widely used microbially produced biosurfactants are glycolipids. These compounds have distinct hydrophilic and hydrophobic regions, and the final compound structure and characteristics depend on the particular growth conditions and the carbon source used. Good yields often are obtained with insoluble substrates. These biosurfactants are excellent dispersing agents and have been used with the *Exxon Valdez* oil spill.

Bioconversion Processes

Bioconversions, also known as **microbial transformations** or **biotransformations**, are minor changes in molecules, such as the insertion of a hydroxyl or keto function or the saturation/desaturation of a complex cyclic structure, that are carried out by nongrowing microorganisms. The microorganisms thus act as **biocatalysts**. Bioconversions have many advantages over chemical procedures. A major advantage is stereochemical; the biologically active form of a product is made. In contrast, most chemical syntheses produce racemic mixtures in which only one of the two isomers will be able to be used efficiently by the organism. Enzymes also carry out very specific reactions under mild conditions, and larger water-insoluble molecules can be transformed. Unicellular bacteria, actinomycetes, yeasts, and molds have been used in various bioconversions. The enzymes responsible for these conversions can be intracellular or extracellular. Cells can be produced in batch or continuous culture and then dried for direct use, or they can be prepared in more specific ways to carry out desired bioconversions.

A typical bioconversion is the hydroxylation of a steroid (**figure 42.14**). In this example, the water-insoluble steroid is dissolved in acetone and then added to the reaction system that contains the pregrown microbial cells. The course of the modification is monitored, and the final product is extracted from the medium and purified.

Biotransformations carried out by free enzymes or intact nongrowing cells do have limitations. Reactions that occur in the absence of active metabolism—without reducing power or ATP being available continually—are primarily exergonic reactions (see section 8.3). If ATP or reductants are required, an energy

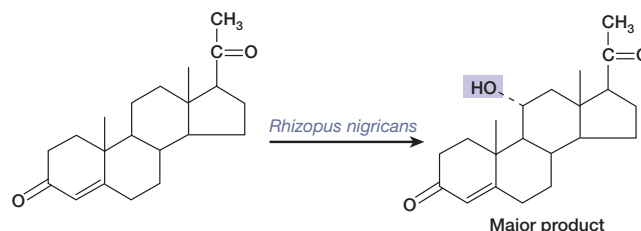


Figure 42.14 Biotransformation to Modify a Steroid.

Hydroxylation of progesterone in the 11 α position by *Rhizopus nigricans*. The steroid is dissolved in acetone before addition to the pregrown fungal culture.

source such as glucose must be supplied under carefully controlled nongrowth conditions.

When freely suspended vegetative cells or spores are employed, the microbial biomass usually is used only once. At the end of the process, the cells are discarded. Cells often can be used repeatedly after attaching them to ion exchange resins by ionic interactions or immobilizing them in a polymeric matrix. Ionic, covalent, and physical entrapment approaches can be used to immobilize microbial cells, spores, and enzymes. Microorganisms also can be immobilized on the inner walls of fine tubes. The solution to be modified is then simply passed through the microorganism-lined tubing; this approach is being applied in many industrial and environmental processes. These include bioconversions of steroids, degradation of phenol, and the production of a wide range of antibiotics, enzymes, organic acids, and metabolic intermediates. One application of cells as biocatalysts is the recovery of precious metals from dilute-process streams.

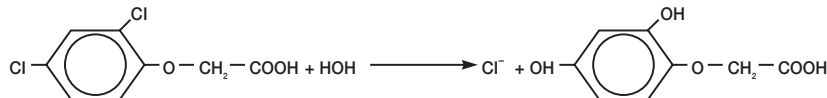
1. Discuss the major uses for biopolymers and biosurfactants.
2. What are cyclodextrins and why are they important additives?
3. What are bioconversions or biotransformations? Describe the changes in molecules that result from these processes.

42.4 Microbial Growth in Complex Environments

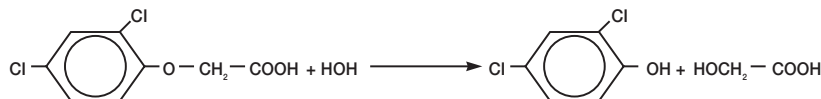
Industrial microbiology and biotechnology also can be carried out in complex natural environments such as waters, soils, or high organic matter-containing composts. In these complex environments, the physical and nutritional conditions for microbial growth cannot be completely controlled, and a largely unknown microbial community is present. These applications of industrial microbiology and biotechnology usually are lower cost, larger volume processes, where no specific commercial microbial product is created. Examples are (1) the use of microbial communities to carry out biodegradation, bioremediation, and environmental maintenance processes; and (2) the addition of microorganisms to soils or plants for the improvement of crop production. Both of these applications will be discussed in this section.

Figure 42.15 Biodegradation Has Several Meanings. Biodegradation is a term that can be used to describe three major types of changes in a molecule. (a) A minor change in the functional groups attached to an organic compound, as the substitution of a hydroxyl group for a chlorine group. (b) An actual breaking of the organic compound into organic fragments in such a way that the original molecule could be reconstructed. (c) The complete degradation of an organic compound to minerals.

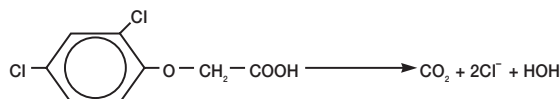
(a) Minor change (dehalogenation)



(b) Fragmentation



(c) Mineralization



Biodegradation Using Natural Microbial Communities

Before discussing **biodegradation** processes carried out by natural microbial communities, it is important to consider definitions. Biodegradation has at least three definitions (**figure 42.15**): (1) a minor change in an organic molecule leaving the main structure still intact, (2) fragmentation of a complex organic molecule in such a way that the fragments could be reassembled to yield the original structure, and (3) complete mineralization. As mentioned previously (*see p. 613*), mineralization is the transformation of organic molecules to mineral forms, including carbon dioxide or methane, plus inorganic forms of other elements that might have been contained in the original structures.

Originally it was assumed, given time and the almost infinite variety of microorganisms, that all organic compounds, including those synthesized in the laboratory, would eventually degrade. Observations of natural and synthetic organic compound accumulation in natural environments, however, began to raise questions about the ability of microorganisms to degrade these varied substances and the role of the environment (clays, anaerobic conditions) in protecting some chemicals. With the development of synthetic pesticides, it became distressingly evident that not all organic compounds are immediately biodegradable. This chemical **recalcitrance** (resisting authority or control) resulted from the apparent fallibility of microorganisms, or their inability to degrade some industrially synthesized chemical compounds.

Degradation of a complex compound takes place in several stages. In the case of halogenated compounds, dehalogenation often occurs early in the overall process. Dehalogenation of many compounds containing chlorine, bromine, or fluorine occurs faster under anaerobic than under aerobic conditions. The study of **reductive dehalogenation**, especially its commercial applications, is expanding rapidly. Research on the dehalogenation of PCBs shows that this coreductive process can use electrons de-

rived from water; other studies indicate that hydrogen can be the source of reductant for the dehalogenation of different chlorinated compounds. Major genera that carry out this process include *Desulfitobacterium*, *Dehalospirillum*, and *Desulfomonile*.

Humic acids, brownish polymeric residues of lignin decomposition that accumulate in soils and waters, have been found to play a role in anaerobic biodegradation processes. They can serve as electron acceptors under what are called “humic-acid-reducing conditions.” The use of humic acids as electron acceptors has been observed with the anaerobic dechlorination of vinyl chloride and dichloroethylene.

Once the anaerobic dehalogenation steps are completed, degradation of the main structure of many pesticides and other xenobiotics often proceeds more rapidly in the presence of O_2 .

Structure and stereochemistry are critical in predicting the fate of a specific chemical in nature. When a constituent is in the meta as opposed to the ortho position, the compound will be degraded at a much slower rate. The **meta effect** is shown in **figure 42.16**. This stereochemical difference is the reason that the common lawn herbicide 2,4-dichlorophenoxyacetic acid (2,4-D), with a chlorine in the ortho position, will be largely degraded in a single summer. In contrast, 2,4,5-trichlorophenoxyacetic acid, with a constituent in the meta position, will persist in the soils for several years, and thus is used for long-term brush control. Check out the labels on herbicide preparations the next time you go to the garden store!

An important aspect of managing biodegradation is the recognition that many of the compounds that are added to environments are **chiral**, or possess asymmetry and handedness. Microorganisms often can degrade only one isomer of a substance; the other isomer will remain in the environment. At least 25% of herbicides are chiral (**figure 42.17**). Thus it is critical to add the herbicide isomer that is effective and also degradable. Recent studies have shown that microbial communities in different environments will degrade dif-

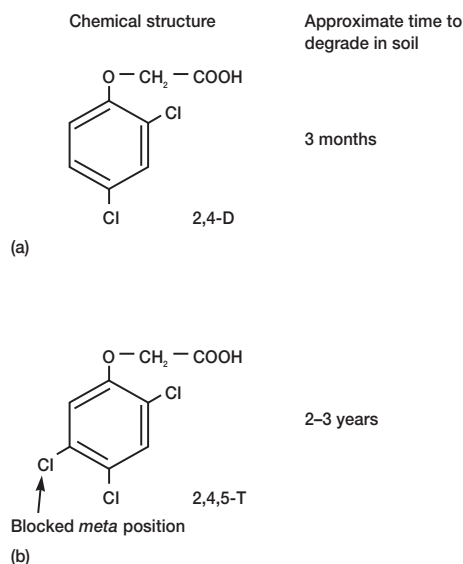


Figure 42.16 The Meta Effect and Biodegradation. Minor structural differences can have major effects on the biodegradability of chemicals. The *meta* effect is an important example. (a) Readily degradable 2,4-dichlorophenoxyacetic acid (2,4-D) with an exposed *meta* position on the ring degrades in several months; (b) recalcitrant 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) with the blocked *meta* group, can persist for years.

ferent enantiomers. Changes in environmental conditions and nutrient supplies can alter the patterns of chiral form degradation.

Microbial communities change their characteristics in response to physical changes such as mixing of soil or water to add oxygen, or after the addition of inorganic or organic substrates, which may stimulate different components of the microbial community. If a particular compound, such as a herbicide, is added repeatedly to a microbial community, the community adapts and faster rates of degradation can occur (figure 42.18). The adaptive process often is so effective that this enrichment culture-based approach, established on the principles elucidated by Beijerinck (*see p. 11*) can be used to isolate organisms with a desired set of capabilities. For example, a microbial community can become so efficient at rapid herbicide degradation that herbicide effectiveness is diminished. To counteract this process, herbicides can be changed to throw the microbial community off balance, thus preserving the effectiveness of the chemicals. The degradation of many pesticides may also result in the accumulation of organic fragments that bind with organic matter in the soil. The longer-term fate and possible effects of “bound” pesticide residues on the soil system, plants, and higher organisms are largely unknown.

Degradation processes that occur in soils also can be used in large-scale degradation of hydrocarbon wastes or of wastewater, particularly from agricultural operations, in a technique called **land farming**. The waste material is incorporated into the soil or allowed to flow across the soil surface, where degradation occurs.

It is important to emphasize that such degradation processes do not always reduce environmental problems. In fact, the partial degra-

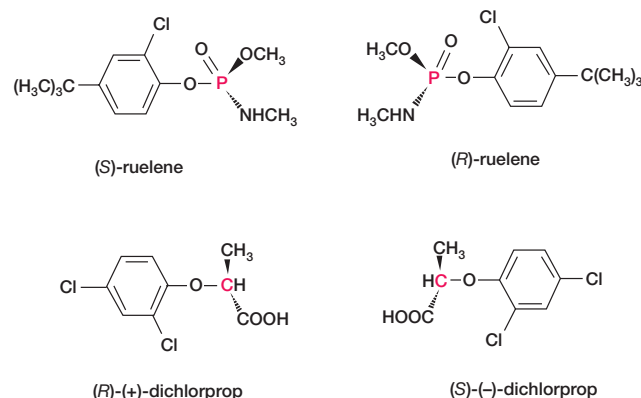


Figure 42.17 Chirality, or Handedness, Is Important in Degradation. It is now recognized that one enantiomer form of a chemical may be more effective, and also may differ in degradability. Enantiomers of the herbicides rueiene and dichlorprop are shown. It is critical to add the isomers that are effective and biodegradable.

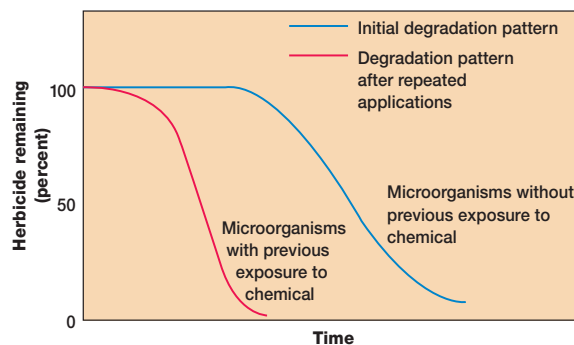
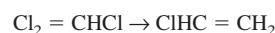


Figure 42.18 Repeated Exposure and Degradation Rate. Addition of an herbicide to a soil can result in changes in the degradative ability of the microbial community. Relative degradation rates for an herbicide after initial addition to a soil, and after repeated exposure to the same chemical.

dation or modification of an organic compound may not lead to decreased toxicity. An example of this process is the microbial metabolism of 1,1,1-trichloro-2,2-bis-(p-chlorophenyl)ethane (DDT), a xenobiotic or foreign (chemically synthesized) organic compound. Degradation removes a chlorine function to give 1,1-dichloro-2,2-bis(p-chlorophenyl)ethylene (DDE), which is still of environmental concern. Another important example is the degradation of trichloroethylene (TCE), a widely used solvent. If this is degraded under anaerobic conditions, the dangerous carcinogen vinyl chloride can be synthesized.



Biodegradation also can lead to widespread damages and financial losses. Metal corrosion is a particularly important example.

Box 42.2

Methanogens: A New Role for an Ancient Microbial Group

The methanogens, an important group of the archaea that can produce methane, are considered to be at least 3.5 billion years old. Despite intensive research, new discoveries are still being made concerning these microorganisms. Methanogens have now been found to contribute to the anaerobic corrosion of soft iron. Previously the microbial group usually considered the major culprit in the anaerobic corrosion process was the genus *Desulfovibrio*, which can use sulfate as an oxidant and hydrogen produced in the corrosion process as a reductant. Methanogens can use elemental iron as an electron source in

their metabolism. It appears that corrosion may occur even without the presence of sulfate, which is required for functioning of *Desulfovibrio*. Rates of iron removal by the methanogens are around 79 mg/1,000 cm² of surface area in a 24 hour period. This may not seem a high rate, but in relation to the planned service life of metal structures in muds and subsurface soils—possibly years and decades—such corrosion can become a major problem. Continuous efforts to improve protection of iron structures will be required in view of the diversity of iron-corroding microorganisms.

The microbially mediated corrosion of metals is particularly critical where iron pipes are used in waterlogged anaerobic environments or in secondary petroleum recovery processes carried out at older oil fields. In these older fields water is pumped down a series of wells to force residual petroleum to a central collection point. If the water contains low levels of organic matter and sulfate, anaerobic microbial communities can develop in rust blebs or tubercles (figure 42.19), resulting in punctured iron pipe and loss of critical pumping pressure. Microorganisms that use elemental iron as an electron donor during the reduction of CO₂ in methanogenesis have recently been discovered (Box 42.2). Because of the wide range of interactions that occur between microorganisms and metals, the need to develop strategies to deal with corrosion problems is critical.

1. Give alternative definitions for the term biodegradation.
2. What is reductive dehalogenation? Describe humic acids and the role they can play in anaerobic degradation processes.
3. Discuss chirality and its importance for understanding degradation effects in the environment.
4. Why is the “meta effect” important for understanding biodegradation?
5. What is “land farming” and why is it important in waste degradation?

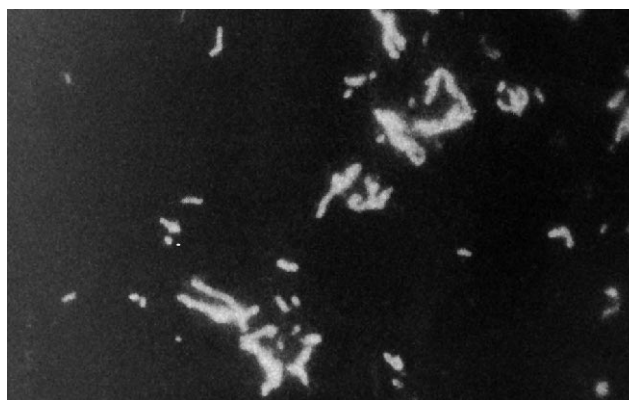
Changing Environmental Conditions
to Stimulate Biodegradation

Often natural microbial communities will not be able to carry out biodegradation processes at a desired rate due to limiting physical or nutritional factors. For example, biodegradation often will be limited by low oxygen levels. Hydrocarbons, nitrogen, phosphorus, and other needed nutrients also may be absent or available only at slow flux rates, thus limiting rates of degradation. In these cases, it is necessary to determine the limiting factors, based on Liebig's and Shelford's laws, and then to supply needed materials or modify the environment. [Liebig's and Shelford's laws \(p. 131\)](#)

Most of the early efforts to stimulate the degradative activities of microorganisms involved the modification of waters and soils by the addition of oxygen or nutrients, now called **engineered biore-**



(a)



(b)

Figure 42.19 Microbial-Mediated Metal Corrosion. The microbiological corrosion of iron is a major problem. (a) The graphitization of iron under a rust bleb on the pipe surface allows microorganisms, including *Desulfovibrio*, to corrode the inner surface. (b) Evidence points to the importance of communities of microorganisms, as opposed to individual species acting alone, as a major factor in microbiologically influenced corrosion. This epifluorescence microscope view ($\times 1,600$) is of pipeline steel a few hours after colonization by sulfate-reducing and organic acid-producing bacteria such as species of *Enterobacter* and *Clostridium*.

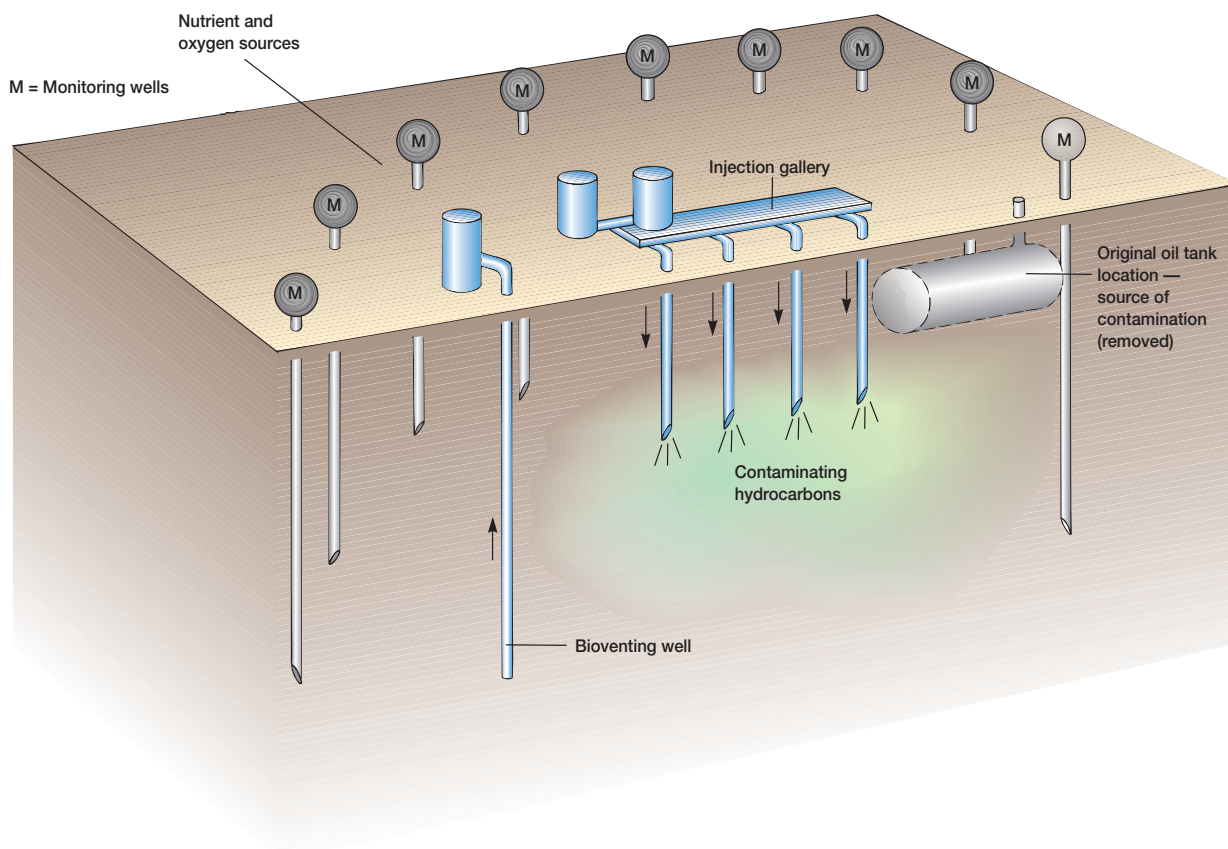


Figure 42.20 A Subsurface Engineered Bioremediation System. Monitoring and recovery wells are used to monitor the plume and its possible movement. Nutrients and oxygen (as peroxide or air) are added to the contaminated soil and groundwater. A bioventing well can be used to accelerate the removal of hydrocarbon vapors.

mediation. Contact between the microbes and the substrate; the proper physical environment, nutrients, oxygen (in most cases); and the absence of toxic compounds are critical in this managed process.

Often it is found that the addition of easily metabolized organic matter such as glucose increases biodegradation of recalcitrant compounds that are usually not used as carbon and energy sources by microorganisms. This process, termed **cometabolism**, is finding widespread applications in biodegradation management. Cometabolism can be carried out by simply adding easily catabolized organic matter such as glucose or cellulose and the compound to be degraded to a complex microbial community. Plants also may be used to provide the organic matter. Cometabolism is important in many different biodegradation systems, and it also is discussed in chapter 30.

Stimulating Hydrocarbon Degradation in Waters and Soils

Experiences with oil spills in marine environments illustrate these principles. When working with dispersed hydrocarbons in the ocean, contact between the microorganism, the hydrocarbon substrate, and other essential nutrients must be maintained. To achieve this, pellets

containing nutrients and an oleophilic (hydrocarbon soluble) preparation have been used. This technique has accelerated the degradation of different crude oil slicks by 30 to 40%, in comparison with control oil slicks where the additional nutrients were not available.

A unique challenge for this technology was the *Exxon Valdez* oil spill, which occurred in Alaska in March 1989. Several different approaches were used to increase biodegradation. These included nutrient additions, chemical dispersants, biosurfactant additions, and the use of high-pressure steam. The use of a microbially produced glycolipid emulsifier has proven helpful.

The degradation of hydrocarbons and other chemical residues in contaminated subsurface environments presents special challenges. The major difference is that geological structures have limited permeability. Although subsurface regions in a pristine state often have O_2 concentrations approaching saturation, the penetration of small amounts of organic matter into these structures can quickly lead to O_2 depletion.

A typical approach that can be used to carry out in situ bioremediation in subsurface environments is shown in **figure 42.20**. Depending on the petroleum contamination and the geological

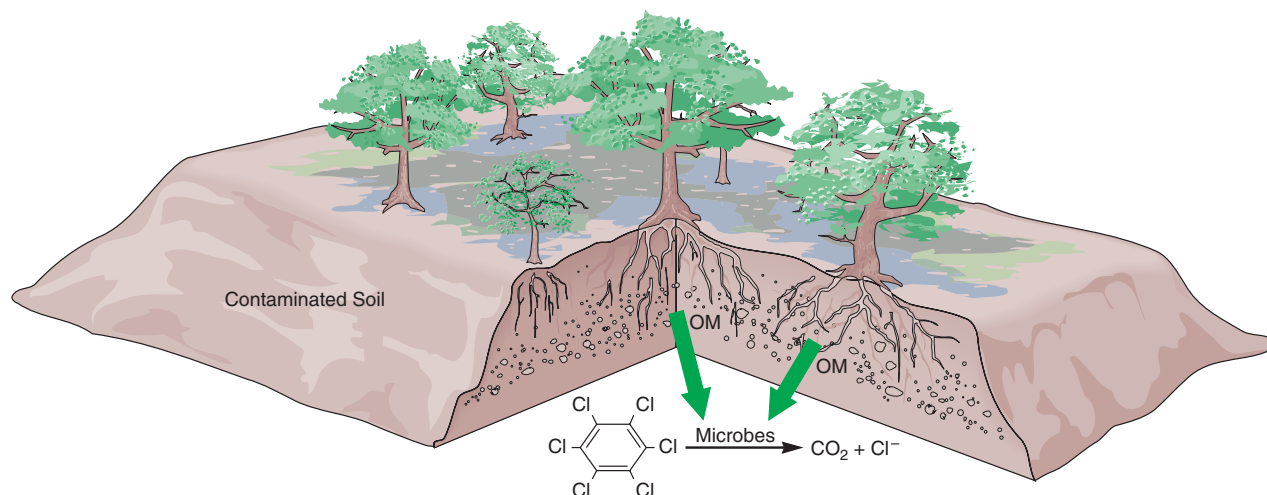


Figure 42.21 Phytoremediation. A conceptual view of a phytoremediation system, with a cut-away section of the root-soil zone. When organic matter (OM) is released from the plant roots, cometabolic processes can be carried out more efficiently by microbes, leading to enhanced degradation of contaminants. The degradation of hexachlorobenzene is shown as an example.

characteristics of the site, injection and monitoring wells can be installed. Nutrients and a source of oxygen (possibly compressed air or peroxide) also can be added. Often this process is combined with bioventing, the physical removal of vapors by a vacuum. Depending on the volume and the location of the contaminated soil, the process may require months or years to complete.

A unique two-stage process can be used to degrade PCBs in river sediments. First, partial dehalogenation of the PCBs occurs naturally under anaerobic conditions. Then the muds are aerated to promote the complete degradation of the less chlorinated residues produced by this intrinsic bioremediation process (chapter opening figure).

Stimulating Degradation with Plants

Phytoremediation, or the use of plants to stimulate the degradation, transformation, or removal of compounds, either directly or in conjunction with microorganisms, is becoming an important part of biodegradation technology. A plant provides nutrients that allow cometabolism to occur in the plant root zone or rhizosphere (**figure 42.21**). Phytoremediation also includes plant contributions to degradation, immobilization, and volatilization processes, as noted in **table 42.12**. Transgenic plants may be employed in phytoremediation. Using cloning techniques with *Agrobacterium* (see pp. 340, 492–93, 684), the *merA* and *merB* genes have been integrated into a plant (*Arabidopsis thaliana*), thus making it possible to transform extremely toxic organic mercury forms to elemental mercury, which is less of an envi-

Table 42.12 Types of Phytoremediation

Process	Function
Phytoextraction	Use of pollutant-accumulating plants to remove metals or organics from soil by concentrating them in the harvestable plant parts
Phytodegradation	Use of plants and associated microorganisms to degrade organic pollutants
Rhizofiltration	Use of plant roots to absorb and adsorb pollutants, mainly metals, from water and aqueous waste streams
Phytostabilization	Use of plants to reduce the bioavailability of pollutants in the environment
Phytovolatilization	Use of plants to volatilize pollutants

Based on T. Macek; M. Mackova; and J. Kás. 2000. Exploitation of plants for the removal of organics in environmental remediation. *Biotechnol. Adv.* 18:23–34. P. 25.

ronmental hazard. Recently transgenic tobacco plants have been constructed that express tetranitrate reductase, an enzyme from an explosive-degrading bacterium, thereby enabling the transgenic plants to degrade nitrate ester and nitro aromatic explosives. The genetically modified plants grow in solutions of explosives that control plants cannot tolerate. Other plants have been engineered in the same way to degrade trichloroethylene, an environmental contaminant of worldwide concern.

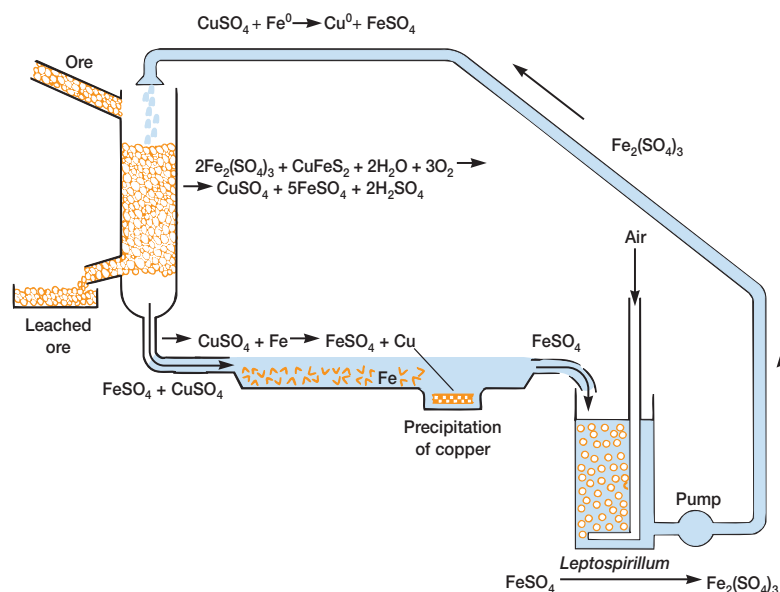
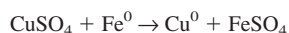


Figure 42.22 Copper Leaching from Low-Grade Ores. The chemistry and microbiology of copper ore leaching involve interesting complementary reactions. The microbial contribution is the oxidation of ferrous ion (Fe^{2+}) to ferric ion (Fe^{3+}). *Leptospirillum ferrooxidans* and related microorganisms are very active in this oxidation. The ferric ion then reacts chemically to solubilize the copper. The soluble copper is recovered by a chemical reaction with elemental iron, which results in an elemental copper precipitate.

Stimulation of Metal Bioleaching from Minerals

Bioleaching is the use of microorganisms, which produce acids from reduced sulfur compounds, to create acidic environments that solubilize desired metals for recovery. This approach is used to recover metals from ores and mining tailings with metal levels too low for smelting. Bioleaching carried out by natural populations of *Leptospirillum*-like species, *Thiobacillus thiooxidans*, and related thiobacilli, for example, allows recovery of up to 70% of the copper in low-grade ores. As shown in **figure 42.22**, this involves the biological oxidation of copper present in these ores to produce soluble copper sulfate. The copper sulfate can then be recovered by reacting the leaching solution, which contains up to 3.0 g/liter of soluble copper, with iron. The copper sulfate reacts with the elemental iron to form ferrosulfate, and the copper is reduced to the elemental form, which precipitates out in a settling trench. The process is summarized in the following reaction:



Bioleaching may require added phosphorus and nitrogen if these are limiting in the ore materials, and the same process can be used to solubilize uranium.

It is apparent that nature will assist in bioremediation if given a chance. The role of natural microorganisms in biodegradation is now better appreciated. An excellent example is the recent work with the very versatile fungus *Phanerochaete chrysosporium* (**Box 42.3**).

Often biodegradation and biodeterioration have major negative effects, and it becomes important to control and limit these processes by environmental management. Problems include unwanted degradation of paper, jet fuels, textiles, and leather goods. A global concern is microbial-based metal corrosion.

1. What factors must one consider when attempting to stimulate the microbial degradation of a massive oil spill in a marine environment?
2. What is cometabolism and why is this important for degradation processes?
3. How is in situ bioremediation carried out?
4. Describe the major types of phytoremediation. What is the role of microorganisms in each of these processes?
5. How is bioleaching carried out and what microbial genera are involved?
6. What is unique about *Phanerochaete chrysosporium*? What does its name mean?

Addition of Microorganisms to Complex Microbial Communities

Both in laboratory and field studies, attempts have been made to speed up existing microbiological processes by adding known active microorganisms to soils, waters, or other complex systems. The microbes used in these experiments have been isolated from contaminated sites, taken from culture collections, or derived from uncharacterized enrichment cultures. For example, commercial culture preparations are available to facilitate silage formation and to improve septic tank performance.

Addition of Microorganisms without Considering Protective Microhabitats

With the development of the “superbug” by A. M. Chakrabarty in 1974, there was initial excitement due to the hope that such

Box 42.3

Phanerochaete chrysosporium: A Wood-Degrading Fungus with a Voracious Appetite

The basidiomycete *Phanerochaete chrysosporium* (the scientific name means “visible hair, golden spore”) is a fungus with unusual degradative capabilities. This organism is termed a “white rot fungus” because of its ability to degrade lignin, a randomly linked phenylpropane-based polymeric component of wood (see section 28.3). The cellulosic portion of wood is attacked to a lesser extent, resulting in the characteristic white color of the degraded wood. This organism also degrades a truly amazing range of xenobiotic compounds (nonbiological foreign chemicals) using both intracellular and extracellular enzymes.

As examples, the fungus degrades benzene, toluene, ethylbenzene, and xylenes (the so-called BTEX compounds), chlorinated compounds such as 2,4,5-trichloroethylene (TCE), and trichlorophenols. The latter are present as contaminants in wood preservatives and also are used as pesticides. In addition, other chlorinated benzenes can be degraded with or without toluenes being present. Even the insecticide Hydramethylnon is degraded!

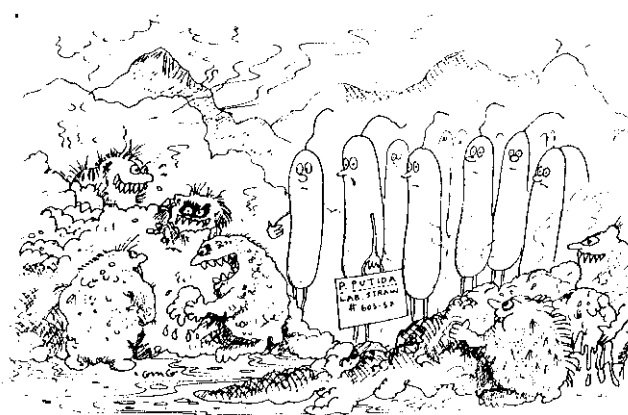
How does this microorganism carry out such feats? Apparently most degradation of these xenobiotic compounds occurs after active growth,

during the secondary metabolic lignin degradation phase. Degradation of some compounds involves important extracellular enzymes including lignin peroxidase, manganese-dependent peroxidase, and glyoxal oxidase. A critical enzyme is pyranose oxidase, which releases H_2O_2 for use by the manganese-dependent peroxidase enzyme. The H_2O_2 also is a precursor of the highly reactive hydroxyl radical, which participates in wood degradation. Apparently the pyranose oxidase enzyme is located in the interperiplasmic space of the fungal cell wall, where it can function either as a part of the fungus or be released from the fungus and penetrate into the wood substrate. It appears that the nonspecific enzymatic system that releases these oxidizing products degrades many cyclic, aromatic, and chlorinated compounds related to lignins.

We can expect to continue hearing of many new advances in work with this organism. Potentially valuable applications being studied include growth in bioreactors where intracellular and extracellular enzymes can be maintained in the bioreactor while liquid wastes flow past the immobilized fungi.

an improved microorganism might be able to degrade hydrocarbon pollutants very effectively. A critical point, which was not considered, was the actual location, or microhabitat, where the microbe had to survive and function. Engineered microorganisms were added to soils and waters with the expectation that rates of degradation would be stimulated as these microorganisms established themselves. Generally such additions led to short-term increases in rates of the desired activity, but typically after a few days the microbial community responses were similar in treated and control systems. After many unsuccessful attempts, it was found that the lack of effectiveness of such added cultures was due to at least three factors: (1) the attractiveness of laboratory-grown microorganisms as a food source for predators such as soil protozoa, (2) the inability of these added microorganisms to contact the compounds to be degraded, and (3) the failure of the added microorganisms to survive and compete with indigenous microorganisms (figure 42.23). Such a modified microorganism may be less fit to compete and survive because of the additional energetic burden required to maintain the extra DNA.

Attempts have been made to make such laboratory-grown cultures more capable of survival in a natural environment by growing them in low-nutrient media or starving the microorganisms before adding them to an environment. These “toughening” approaches have improved microbial survival and function somewhat, but have not solved the problem. In recent years, there has been less interest in simply adding microorganisms to environments without consid-



“Oh dear! I didn’t realize ‘in the field’ would be like this!
We should have stayed in the laboratory.”

Figure 42.23 A Cartoonist’s View of Laboratory-Grown Microbes Returning to Their Original Environment.

Source: Tibtech 1993 11:344–352.

ering the specific niche or microenvironment in which they are to survive and function. This has led to the field of **natural attenuation**, which emphasizes the use of natural microbial communities in the environmental management of pollutants.

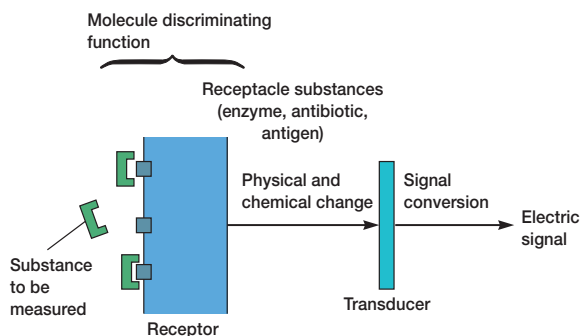


Figure 42.24 Biosensor Design. Biosensors are finding increasing applications in medicine, industrial microbiology, and environmental monitoring. In a biosensor a biomolecule or whole microorganism carries out a biological reaction, and the reaction products are used to produce an electrical signal.

Addition of Microorganisms Considering Protective Microhabitats

Microorganism additions to natural environments can be more successful if the microorganism is added together with a microhabitat that gives the organism physical protection, as well as possibly supplying nutrients. This makes it possible for the microorganism to survive in spite of the intense competitive pressures that exist in the natural environment, including pressure from protozoan predators such as ciliates, flagellates, and amoebae. Microhabitats may be either living or inert. [Predation \(pp. 607–9\)](#)

Living Microhabitats. Specialized living microhabitats include the surface of a seed, a root, or a leaf, which, with their higher nutrient flux rate and the chance for initial colonization by the added microorganisms, can protect the added microbe from the fierce competitive conditions in the natural environment. Examples include the use of *Rhizobium* and *Bacillus thuringiensis*. In order to ensure that *Rhizobium* is in close association with the legume, seeds are coated with the microbe using an oil-organism mixture, or *Rhizobium* is placed in a band under the seed where the newly developing primary root will penetrate. In contrast, *Bacillus thuringiensis* (BT) is placed on the surface of the plant leaf, or the plant is engineered to contain the BT genes that allow the production of the toxic protein in situ, once it is ingested. After ingestion by the target organism, the toxic protein will be within the digestive tract where it is most effective. [Bacillus thuringiensis \(pp. 525, 1020–21\); Rhizobium \(sections 22.1 and 30.4\)](#)

Inert Microhabitats. Recently it has been found that microorganisms can be added to natural communities together with protective inert microhabitats! As an example, if microbes are added to a soil with microporous glass, the survival of added microorganisms can be markedly enhanced. Other microbes have been

Table 42.13 Biosensors: Potential Biomedical, Industrial, and Environmental Applications

Clinical diagnosis and biomedical monitoring
Agricultural, horticultural, veterinary analysis
Detection of pollution, and microbial contamination of water
Fermentation analysis and control
Monitoring of industrial gases and liquids
Measurement of toxic gas in mining industries
Direct biological measurement of flavors, essences, and pheromones

observed to create their own microhabitats! Microorganisms in the water column overlying PCB-contaminated sand-clay soils have been observed to create their own “clay hutches” by binding clays to their outer surfaces with exopolysaccharides. These illustrations show that with the application of principles of microbial ecology it may be possible to more successfully manage microbial communities in nature.

1. What factors might limit the ability of microorganisms, after addition to a soil or water, to be able to persist and carry out desired functions?
2. What types of microhabitats can be used with microorganisms when they are added to a complex natural environment?
3. Why are plants inoculated with *Bacillus thuringiensis*?

42.5 Biotechnological Applications

Microorganisms and parts of microorganisms, especially enzymes, are used in a wide variety of biotechnological applications to monitor the levels of critical compounds in the environment and in animals and humans. These techniques have wide applications in environmental science, animal and human health, and in basic science.

Biosensors

A rapidly developing area of biotechnology, arousing intense international scientific interest, is that of **biosensor** production. In this field of bioelectronics, living microorganisms (or their enzymes or organelles) are linked with electrodes, and biological reactions are converted into electrical currents by these biosensors (**figure 42.24**). Biosensors are being developed to measure specific components in beer, to monitor pollutants, and to detect flavor compounds in food. It is possible to measure the concentration of substances from many different environments (**table 42.13**). Applications include the detection of glucose, acetic acid, glutamic acid, ethanol, and biochemical oxygen demand. In addition, the application of biosensors

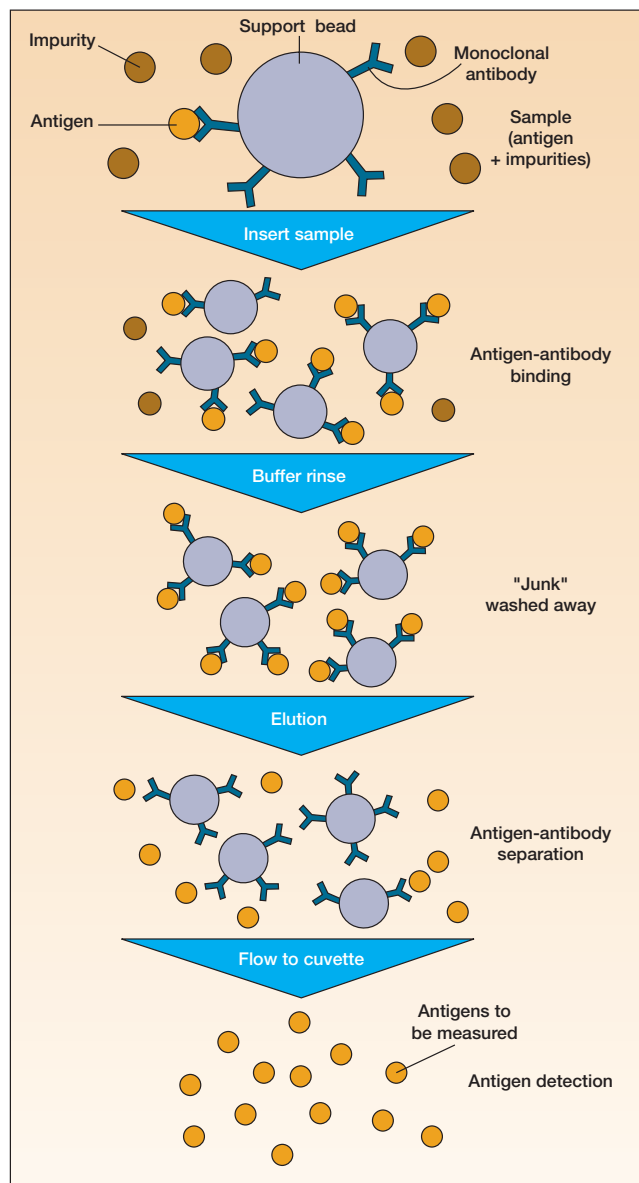


Figure 42.25 A Biosensor for Rapid Detection of a Pathogen. Basic reaction scheme for the immunochemical-based capture, purification, and detection of a pathogen based on a monoclonal antibody system. Detection can be carried out using a small portable instrument.

to measure cephalosporin, nicotinic acid, and several B vitamins has been described. Recently biosensors have been developed using immunochemical-based detection systems (**figure 42.25**). These new biosensors will detect pathogens, herbicides, toxins, proteins, and DNA. Many of these biosensors are based on the use of a streptavidin-biotin recognition system (**Box 42.4**).

One of the most interesting recent developments using these approaches is a handheld aflatoxin detection system for use in monitoring food quality. This automated unit, based on a new column-based immunoaffinity fluorometric procedure, can be used for 100 measurements before being recharged. The unit can detect from 0.1 to 50 ppb of aflatoxins in a 1.0 ml sample in less than 2 min. [Aflatoxins \(pp. 967–68\)](#)

Rapid advances are being made in all areas of biosensor technology. These include major improvements in the stability and durability of these units, which are being made more portable and sensitive. Microorganisms and metabolites such as glucose can be measured, thus meeting critical needs in modern medicine

Microarrays

A large part of the new and developing microbial biotechnology involves the use of DNA sequences in **gene arrays** to monitor gene expression in complex biological systems (*see section 15.6*). The rapid advances that have occurred in this area are the result of progress in genomics, recombinant DNA technology, optics, fluid flow systems, and high-speed data acquisition and processing. This **microarray technology** has been suggested to provide the equivalent of the chemist's periodic table. It offers the potential of assaying all genes used to assemble an organism and can monitor expression of tens of thousands of genes based on the principles shown in **figure 42.26**. In this technique, 100 to 200 μl volumes, containing desired sequences, are spotted onto glass slides or other inert materials and dried. These arrays are then mixing with cDNAs from gene expression (*see p. 321*). Binding of the cDNA for various genes is measured using rapid photometric monitoring techniques. [Genomics \(chapter 15\)](#); [Nucleic acid hybridization \(pp. 431–32\)](#)

Commercial microarray products are now available that contain 6,400 open frames for screening gene expression in *Saccharomyces cerevisiae*. For *E. coli*, 4,200 open reading frames can be scanned in a microarray format. These approaches, both now and in the future, make it possible to follow the expression of thousands of genes and study global regulation of microbial growth and responses to environmental changes.

1. What are biosensors and how do they detect substances?
2. What areas are biosensors being used in to assist in chemical and biological monitoring efforts?
3. Describe streptavidin-biotin systems and how they work. Why is this technique important?
4. What is a gene array? What basic techniques are used in this new procedure?

Biopesticides

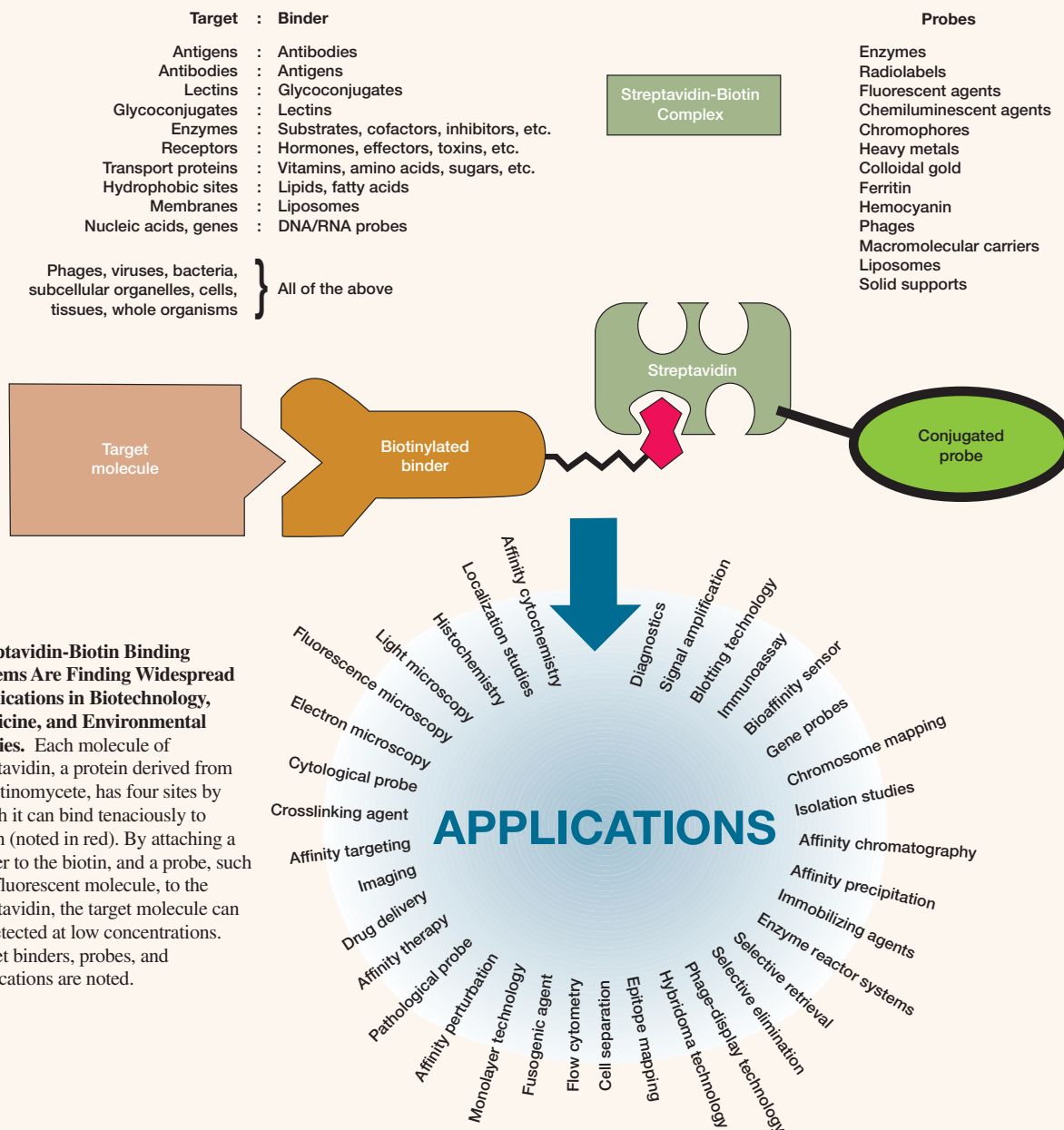
There has been a long-term interest in the use of bacteria, fungi, and viruses as **bioinsecticides** and **biopesticides** (**table 42.14**). These are defined as biological agents, such as bacteria, fungi, viruses, or their components, which can be used to kill a susceptible insect. In this section, major uses of bacteria, fungi, and viruses to control populations of insects will be discussed.

Box 42.4

Streptavidin-Biotin Binding and Biotechnology

Egg white contains many proteins and glycoproteins with unique properties. One of the most interesting, which binds tenaciously to biotin, was isolated in 1963. This glycoprotein, called avidin due to its “avid” binding of biotin, was suggested to play an important role: making egg white antimicrobial by “tying up” the biotin needed by many microorganisms. Avidin, which functions best under alkaline conditions, has the highest known binding affinity between a protein and a ligand. Several years later, scientists at Merck & Co., Inc. discovered a similar protein produced by an actinomycete, *Streptomyces avidini*, which binds biotin at a neutral pH and which does not contain carbohydrates. These characteristics make streptavidin an ideal binding agent for biotin, and it has

been used in an almost unlimited range of applications, as shown in the **Box figure**. The streptavidin protein is joined to a probe. When a sample is incubated with the biotinylated binder, the binder attaches to any available target molecules. The presence and location of target molecules can be determined by treating the sample with a streptavidin probe because the streptavidin binds to the biotin on the biotinylated binder, and the probe is then visualized. This detection system is being employed in a wide variety of biotechnological applications, including use as a nonradioactive probe in hybridization studies and as a critical component in biosensors for a wide range of environmental monitoring and clinical applications. Not bad for a protein from a “simple” filamentous bacterium!



Streptavidin-Biotin Binding Systems Are Finding Widespread Applications in Biotechnology, Medicine, and Environmental Studies. Each molecule of streptavidin, a protein derived from an actinomycete, has four sites by which it can bind tenaciously to biotin (noted in red). By attaching a binder to the biotin, and a probe, such as a fluorescent molecule, to the streptavidin, the target molecule can be detected at low concentrations. Target binders, probes, and applications are noted.

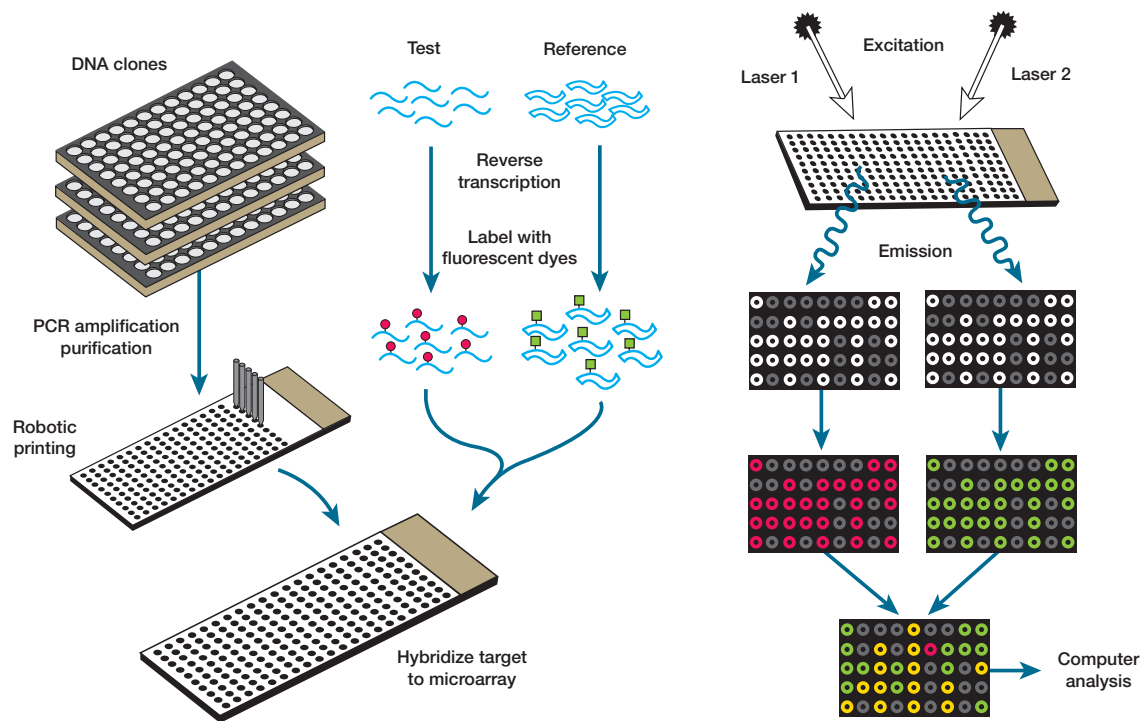


Figure 42.26 A Microarray System for Monitoring Gene Expression. Cloned genes from an organism are amplified by PCR, and after purification, samples are placed on a support in a pattern using a robotic printer. To monitor enzyme expression, RNA from test and reference cultures are converted to cDNA by a reverse transcriptase and labeled with two different fluor dyes. The labeled mixture is hybridized to the microarray and scanned using two lasers with different exciting wavelengths. After pseudocoloring, the fluorescence responses are measured as normalized ratios that show whether the test gene response is higher or lower than that of the reference.

Table 42.14 The Use of Bacteria, Viruses, and Fungi As Bioinsecticides:
An Older Technology with New Applications

Microbial Group	Major Organisms and Applications
Bacteria	<i>Bacillus thuringiensis</i> and <i>Bacillus popilliae</i> are the two major bacteria of interest. <i>Bacillus thuringiensis</i> is used on a wide variety of vegetable and field crops, fruits, shade trees, and ornamentals. <i>B. popilliae</i> is used primarily against Japanese beetle larvae. Both bacteria are considered harmless to humans. <i>Pseudomonas fluorescens</i> , which contains the toxin-producing gene from <i>B. thuringiensis</i> , is used on maize to suppress black cutworms.
Viruses	Three major virus groups that do not appear to replicate in warm-blooded animals are used: nuclear polyhedrosis virus (NPV), granulosis virus (GV), and cytoplasmic polyhedrosis virus (CPV). These occluded viruses are more protected in the environment.
Fungi	Over 500 different fungi are associated with insects. Infection and disease occur primarily through the insect cuticle. Four major genera have been used. <i>Beauveria bassiana</i> and <i>Metarhizium anisopliae</i> are used for control of the Colorado potato beetle and the froghopper in sugarcane plantations, respectively. <i>Verticillium lecanii</i> and <i>Entomophthora</i> spp., have been associated with control of aphids in greenhouse and field environments.

Bacteria

Bacterial agents include a variety of *Bacillus* species, primarily *B. thuringiensis* (see p. 525). This bacterium is only weakly toxic to insects as a vegetative cell, but during sporulation, it produces an

intracellular protein toxin crystal, the parasporal body, that can act as a microbial insecticide for specific insect groups.

The parasporal crystal, after exposure to alkaline conditions in the hindgut, fragments to release the protoxin. After this reacts with a protease enzyme, the active toxin is released (figure 42.27).

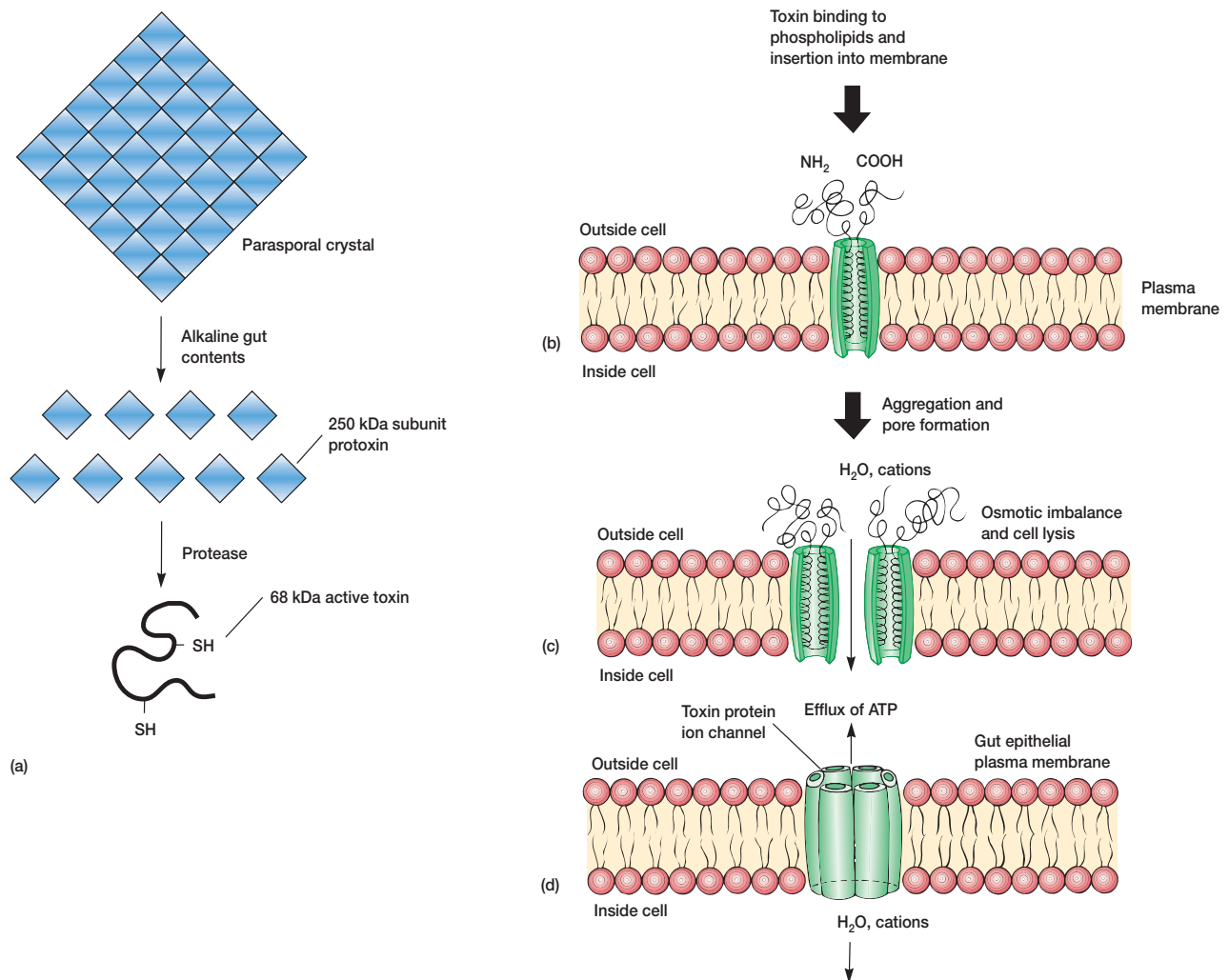


Figure 42.27 The Mode of Action of the *Bacillus thuringiensis* Toxin. (a) Release of the protoxin from the parasporal body and modification by proteases in the hindgut. (b) Insertion of the 68 kDa active toxin molecules into the membrane. (c) Aggregation and pore formation, showing a cross section of the pore. (d) Final creation of the hexagonal pore which causes an influx of water and cations as well as a loss of ATP, resulting in cell imbalance and lysis.

Six of the active toxin units integrate into the plasma membrane (figure 42.27b,c) to form a hexagonal-shaped pore through the midgut cell, as shown in figure 42.27d. This leads to the loss of osmotic balance and ATP, and finally to cell lysis.

The most recent advances in our understanding of *Bacillus thuringiensis* have involved the creation of pest-resistant plants. The first step in this work was to insert the toxin gene into *E. coli*. This work showed that the crystal protein could be expressed in another organism, and that the toxin was effective. This major scientific advance was followed in 1987 by the production of tomato plants that contained the toxin gene.

B. thuringiensis can be grown in fermenters. When the cells lyse, the spores and crystals are released into the medium. The

medium is then centrifuged and made up as a dust or wettable powder for application to plants.

A related bacterium, *Bacillus popilliae*, is used to combat the Japanese beetle. This bacterium, however, cannot be grown in fermenters, and inocula must be grown in the living host. The microorganism controls development of larvae, but destruction of the adult beetle requires chemical insecticides.

Viruses

Viruses that are pathogenic for specific insects include nuclear polyhedrosis viruses (NPVs), granulosis viruses (GVs), and cytoplasmic polyhedrosis viruses (CPVs). Currently over 125 types of NPVs are

known, of which approximately 90% affect the *Lepidoptera*—butterflies and moths. Approximately 50 GVs are known, and they, too, primarily affect butterflies and moths. CPVs are the least host-specific viruses, affecting about 200 different types of insects. An important commercial viral pesticide is marketed under the trade name Elcar for control of the cotton bollworm *Heliothis zea*.

One of the most exciting advances involves the use of baculoviruses that have been genetically modified to produce a potent scorpion toxin active against insect larvae. After ingestion by the larvae, viruses are dissolved in the midgut and are released. Because the recombinant baculovirus produces this insect-selective neurotoxin, it acts more rapidly than the parent virus, and leaf damage by insects is markedly decreased. [Characteristics of insect viruses \(p. 415\)](#)

Fungi

Fungi also can be used to control insect pests. Fungal bioinsecticides, as listed in table 42.14, are finding increasing use in agriculture. The development of biopesticides is progressing rapidly.

Available bioinsecticides which are derived from fungi include kasugamycin and the polyoxins; in addition, special microbiological metabolites such as nikkomycin and the spinosyns are active against insects.

1. What two important bacteria have been used as bioinsecticides?
2. Briefly describe how the *Bacillus thuringiensis* toxin kills insects.
3. What types of viruses are being used to attempt to control insects? What is a trade name for one of these products?
4. Which fungi presently are being used as biopesticides?

42.6 Impacts of Microbial Biotechnology

The use of microorganisms in industrial microbiology and biotechnology, as discussed in this chapter, does not take place in an ethical and ecological vacuum. Decisions to make a particular

product, and also the methods used, can have long-term and often unexpected effects, as with the appearance of antibiotic-resistant pathogens around the world.

Microbiology is a critical part of the area of **industrial ecology**, concerned with tracking the flow of elements and compounds through the natural and social worlds, or the **biosphere** and the **anthrosphere**. Microbiology, especially as an applied discipline, should be considered within its supporting social world.

Microorganisms have been of immense benefit to humanity through their role in food production and processing, the use of their products to improve human and animal health, in agriculture, and for the maintenance and improvement of environmental quality. Other microorganisms, however, are important pathogens and agents of spoilage, and microbiologists have helped control or limit the activities of these harmful microorganisms. The discovery and use of beneficial microbial products, such as antibiotics, have contributed to a doubling of the human life span in the last century.

A microbiologist who works in any of these areas of biotechnology should consider the longer-term impacts of possible technical decisions. An excellent introduction to the relationship between technology and possible societal impacts is given by Samuel Florman (see Additional Reading). Our first challenge, as microbiologists, is to understand, as much as is possible, the potential impacts of new products and processes on the broader society as well as on microbiology. An essential part of this responsibility is to be able to communicate effectively with the various “societal stakeholders” about the immediate and longer-term potential impacts of microbial-based (and other) technologies.

1. Discuss possible ethical and ecological impacts of a particular product or process discussed in this chapter. Think in terms of the broadest possible impacts in your discussion of this problem.
2. Define industrial ecology.
3. What are the biosphere and anthrosphere? Why might you think the term anthrosphere was coined?

Summary

1. Industrial microbiology has been used to manufacture such products as antibiotics, amino acids, and organic acids and has had many important positive effects on animal and human health. Most work in this area has been carried out using microorganisms isolated from nature or modified by the use of classic mutation techniques. Biotechnology involves the use of molecular techniques to modify and improve microorganisms.
2. Finding new microorganisms in nature for use in biotechnology is a continuing challenge. For most environments, only a very small part of the observable microbial community has been examined (tables 42.1 and 42.2).
3. Selection and mutation continue to be important approaches for identifying new microorganisms. These well-established procedures are now being complemented by molecular techniques, including metabolic engineering and combinatorial biology. With combinatorial biology (table 42.3), it is possible to transfer genes from one organism to another organism, and to form new products (figure 42.5).
4. Site-directed mutagenesis and protein engineering are used to modify gene expression. These approaches are leading to new and often different products with new properties (figure 42.4).
5. Natural genetic engineering is of increasing interest. This involves exploiting microbial responses to stress in adaptive mutation and forced evolution, with the hope of identifying microorganisms with new properties.
6. Microorganisms can be grown in controlled environments of various types using fermenters and other culture systems. If defined constituents are used, growth parameters can be chosen and varied in the course of growing a microorganism. This approach is used particularly for the production of amino acids, organic acids, and antibiotics (figures 42.10 and 42.11).

- Growth in controlled environments is expensive and is used primarily for products employed in maintaining and improving animal and human health.
- Specialty nonantibiotic compounds are an important part of industrial microbiology and biotechnology. These include widely used antitumor agents (table 42.11).
- A wide variety of compounds are produced in industrial microbiology that impact our lives in many ways (table 42.9). These include biopolymers, such as the cyclodextrins (figure 42.13), and biosurfactants. Microorganisms also can be used as biocatalysts to carry out specific chemical reactions (figure 42.14).
- Microorganism growth in complex environments such as soils and waters is not used to create microbial products but to carry out environmental management processes, including bioremediation, plant inoculation, and other related activities. In these cases, the microbes themselves are not final products.
- Biodegradation is a critical part of natural systems mediated largely by microorganisms. This can involve minor changes in a molecule, fragmentation, or mineralization (figure 42.15).
- Biodegradation can be influenced by many factors, including oxygen presence or absence, humic acids, and the presence of readily usable organic matter. Reductive dehalogenation proceeds best under anaerobic conditions, and the presence of organic matter can facilitate modification of recalcitrant compounds in the process of cometabolism.
- The structure of organic compounds influences degradation. If constituents are in specific locations on a molecule, as in the *meta* position (figure 42.16), or if varied structural isomers are present (figure 42.17), degradation can be affected.
- Degradation management can be carried out in place, whether this be large marine oil spills, soils, or the subsurface (figure 42.20). Such large-scale efforts usually involve the use of natural microbial communities.
- Degradation can lead to increased toxicity in many cases. If not managed carefully, widespread pollution can occur. This is particularly critical with land farming, or the spreading of industrial and agricultural wastes on soils to facilitate degradation.
- Plants can be used to stimulate biodegradation processes during phytoremediation. This can involve extraction, filtering, stabilization, and volatilization of pollutants (figure 42.21 and table 42.12).
- Microorganisms can be added to environments that contain complex microbial communities with greater success if living or inert microhabitats are used. These can include living plant surfaces (seeds, roots, leaves) or inert materials such as microporous glass. *Rhizobium* is an important example of a microorganism added to a complex environment using a living microhabitat (the plant root).
- Microorganisms are being used in a wide range of biotechnological applications such as biosensors (figure 42.24). Microarrays are used to monitor gene expression in complex systems (figure 42.26).
- Bacteria, viruses, and fungi can be used as bioinsecticides and biopesticides (table 42.14). *Bacillus thuringiensis* is an important biopesticide, and the BT gene has been incorporated into corn.
- Industrial microbiology and biotechnology can have long-term and possibly unexpected positive and negative effects on the environment, and on animals and humans impacted by these technologies. Advances in biotechnology should be considered in a broad ecological and societal context, which is the focus of industrial ecology.

Key Terms

adaptive mutation 998	engineered bioremediation 1012	non-Newtonian broth 1001
anthrosphere 1022	fermentation 1000	pathway architecture 997
biocatalyst 1009	forced evolution 998	phytoremediation 1014
biodegradation 1010	gene array 1018	primary metabolite 1002
bioinsecticides 1018	industrial ecology 1022	protein engineering 994
biopesticide 1018	land farming 1011	protoplast fusion 994
biopolymer 1007	lyophilization 999	recalcitrance 1010
biosensor 1017	<i>meta</i> effect 1010	reductive dehalogenation 1010
biosphere 1022	metabolic control engineering 997	regulatory mutant 1005
biotransformation 1009	metabolic pathway engineering (MPE) 997	scaleup 1001
chiral 1010	microarray technology 1018	secondary metabolite 1002
combinatorial biology 995	microbial transformation 1009	semisynthetic penicillin 1005
cometabolism 1013	natural attenuation 1016	site-directed mutagenesis 994
continuous feed 1002	natural genetic engineering 998	

Questions for Thought and Review

- What information or technical approaches will be required to be able to characterize the vast majority of microorganisms in nature that have not been grown? Consider that most of these microorganisms are in a resting vegetative state.
- What makes the area of natural genetic engineering unique? Isn't this simply what has been going on in nature since the time microorganisms were first able to function?
- What are the advantages of microarrays for the study of gene expression in complex organisms?
- How is it possible to create a niche or microhabitat for a microorganism? What are the special points of concern in trying to make sure the microbe can find its best place to survive and function?
- How might the "postgenomic" era differ from the "genomic era"?
- Most commercial antibiotics are produced by actinomycetes, and only a few are synthesized by fungi and other bacteria. From physiological and environmental viewpoints, how might you attempt to explain this observation?
- We hear much about the beneficial uses of recombinant DNA technology. What are some of the problems and disadvantages that should be considered when using microorganisms for these applications?

8. Why might *Bacillus thuringiensis* bioinsecticides be of interest in other areas of biotechnology? Consider the molecular aspects of their mode of action.
9. Do you think intrinsic bioremediation can solve all of our environmental pollutant degradation problems? Why or why not?
10. What are some of the possible advantages of biosensors as opposed to more traditional physical and chemical measurement procedures?
11. What are the major types of materials used as nutrients in fermentation media?
12. In what different ways can the term fermentation be used?
13. What parameters can be controlled in a modern industrial fermenter?
14. How do primary and secondary metabolites differ in terms of their synthesis and functions?

Critical Thinking Questions

1. The search for novel plants/microbes and their products can be in direct conflict with the exposure of humans to novel pathogens. Discuss the relative risks and benefits—are there strategies that are more likely to be “win-win”?
2. *Deinococcus radiodurans* is a species of bacteria that is highly resistant to radiation. Can you think of a biotechnological application? How would you test its utility?
3. Discuss the risks of releasing genetically modified microbes or ones that are not natural to the particular environment. What precautions, if any, would you take? What would be your concerns?
4. Why, when a microorganism is removed from a natural environment and grown in the laboratory, will it usually not be able to effectively colonize its original environment if it is grown and added back? Consider the nature of growth media used in the laboratory in comparison to growth conditions in a soil or water when attempting to understand this fundamental problem in microbial ecology.
5. The postgenomic era has been discussed in this and previous chapters of the book. Can you envision the job of a “postgenomicist”?
6. Why is phytoremediation of such current interest for environmental management? Why is it of interest to combine this approach with the use of transgenic plants?
7. The terms biosphere and anthrosphere have been used, together with the term industrial ecology. How does microbial biotechnology relate to these concerns?

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APPENDIX I

A Review of the Chemistry of Biological Molecules

Appendix I provides a brief summary of the chemistry of organic molecules with particular emphasis on the molecules present in microbial cells. Only basic concepts and terminology are presented; introductory textbooks in biology and chemistry should be consulted for a more extensive treatment of these topics.

Atoms and Molecules

Matter is made of elements that are composed of atoms. An element contains only one kind of atom and cannot be broken down to simpler components by chemical reactions. An atom is the smallest unit characteristic of an element and can exist alone or in combination with other atoms. When atoms combine they form molecules. Molecules are the smallest particles of a substance. They have all the properties of the substance and are composed of two or more atoms.

Although atoms contain many subatomic particles, three directly influence their chemical behavior—protons, neutrons, and electrons. The atom's nucleus is located at its center and contains varying numbers of protons and neutrons (**figure AI.1**). Protons have a positive charge, and neutrons are uncharged. The mass of these particles and the atoms that they compose is given in terms of the atomic mass unit (AMU), which is $1/12$ the mass of the most abundant carbon isotope. Often the term dalton (Da) is used to express the mass of molecules. It also is $1/12$ the mass of an atom of ^{12}C or 1.661×10^{-24} grams. Both protons and neutrons have a mass of about one dalton. The atomic weight is the actual measured weight of an element and is almost identical to the mass number for the element, the total number of protons and neutrons in its nucleus. The mass number is indicated by a superscripted number preceding the element's symbol (e.g., ^{12}C , ^{16}O , and ^{14}N).

Negatively charged particles called electrons circle the atomic nucleus (**figure AI.1**). The number of electrons in a neutral atom equals the number of its protons and is given by the atomic number, the number of protons in an atomic nucleus. The atomic number is characteristic of a particular type of atom. For example, carbon has an atomic number of six, hydrogen's number is one, and oxygen's is eight (**table AI.1**).

The electrons move constantly within a volume of space surrounding the nucleus, even though their precise location in this volume cannot

be determined accurately. This volume of space in which an electron is located is called its orbital. Each orbital can contain two electrons. Orbitals are grouped into shells of different energy that surround the nucleus. The first shell is closest to the nucleus and has the lowest energy; it contains only one orbital. The second shell contains four orbitals, one circular and three shaped like dumbbells (**figure AI.2a**). It can contain up to eight electrons. The third shell has even higher energy and holds more than eight electrons. Shells are filled beginning with the innermost

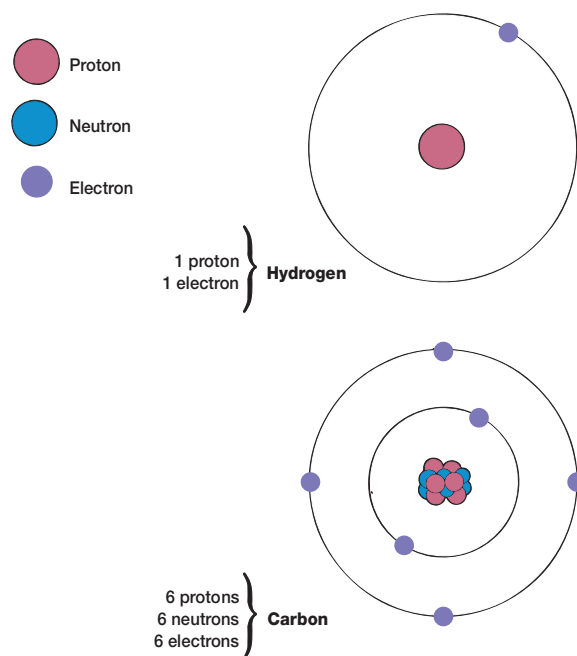


Figure AI.1 Diagrams of Hydrogen and Carbon Atoms. The electron orbitals are represented as concentric circles.

Table AI.1 Atoms Commonly Present in Organic Molecules

Atom	Symbol	Atomic Number	Atomic Weight	Number of Chemical Bonds
Hydrogen	H	1	1.01	1
Carbon	C	6	12.01	4
Nitrogen	N	7	14.01	3
Oxygen	O	8	16.00	2
Phosphorus	P	15	30.97	5
Sulfur	S	16	32.06	2

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